

Targeting poverty over time

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Abstract

Welfare programs that target the poor often update their beneficiary lists infrequently, to facilitate implementation, reduce uncertainty, and minimize targeting costs. Yet standard targeting metrics only consider initial targeting performance, while poverty is notoriously volatile. We develop a framework for evaluating targeting over time, decomposing poverty dynamics into changes in permanent income, transitory shocks, and measurement error, and consider the implications for targeting design. Using household panels from ten developing countries, we apply the framework to Proxy Means Tests (PMTs), often criticized for their inaccuracy. PMTs perform substantially better than the standard approach suggests, and after one year perform as well as targeting those with lowest initial consumption. This is because 80% of poverty dynamics reflect transitory shocks and measurement error, rather than changes in permanent income, so the performance of rules targeting chronic poverty – like PMTs – deteriorates slowly. Consistent with this, in multiple attempts to improve PMTs, via periodic re-targeting, Machine Learning, training on averaged consumption, or predicting future poverty, we find limited gains.

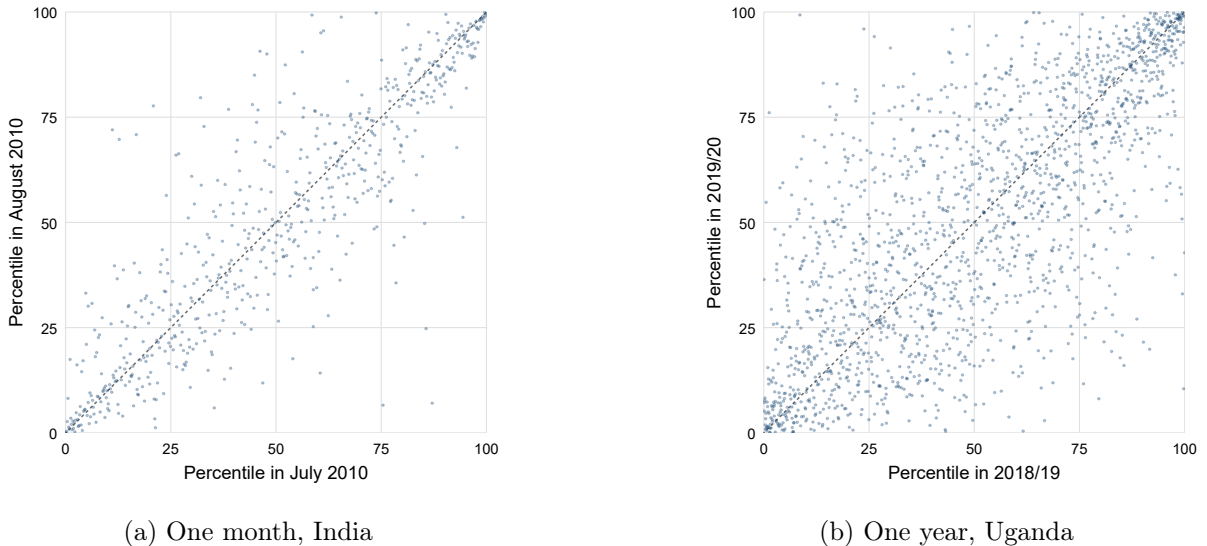
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1. Introduction

Most anti-poverty programs attempt to include the poor and exclude the non-poor, yet do so infrequently. In developing countries, where targeted transfers are a central pillar of social protection for the poorest households, large informal sectors and sparse administrative data make poverty difficult to observe directly. Governments therefore rely on imperfect targeting methods, and the cost of applying those methods—together with the uncertainty and administrative burden created by moving households into and out of programs—means that beneficiary lists are seldom updated. Long-term targeting performance should thus be a key criteria for targeting design. Yet targeting is usually evaluated statically, by asking whether a given rule includes the poor and excludes the non-poor at the time it is calibrated.

While targeting is done infrequently and evaluated statically, observed poverty is notoriously dynamic (Collins et al., 2009; Merfeld and Morduch, 2026; Armentano et al., 2026). Figure 1 shows large movements in relative consumption from one month to the next in India and from one year to the next in Uganda. Translated into a relative poverty line, these movements imply substantial poverty churn: among households initially in the bottom 30 percent, 19.5 percent leave that group after one month in India and 35.8 percent do so after one year in Uganda. This churn raises two immediate questions for targeting. First, how does performance at the time of targeting – the status quo evaluation metric – relate to performance over the lifetime of targeting rules, the welfare-relevant metric? Second, what does this imply for targeting design, and can we hope for infrequent targeting to perform well?

Figure 1 — Churn in (relative) consumption



Notes: These figures plot household percentiles in the observed consumption distribution in one period against the next period. Panel (a) uses monthly consumption data from India. Panel (b) uses annual consumption data from Uganda (LSMS-ISA), which is typical of the data used to train and evaluate targeting performance. The figures show substantial churn in observed relative consumption at both monthly and annual frequency.

In this paper, we develop a framework to study poverty dynamics and its implications

for the evaluation and design of targeting. Linking extensive literatures on targeting and on dynamic consumption, we show that the answers hinge on whether observed poverty dynamics reflect changes in chronic poverty, transient shocks, or measurement error. Measurement error leads status quo evaluation to underestimate performance, with little implied need to retarget over time, while real poverty dynamics lead status quo evaluation to overestimate performance and bound the potential performance of static targeting rules. When dynamics are driven by transient shocks, static rules that target chronic poverty can nevertheless yield stable long-term performance; when dynamics represent changes in chronic poverty, any static targeting method will perform poorly in the long-run.

Applying the framework to Proxy Means Tests (PMTs), among the most widely used targeting methods in practice – and among the most criticized under the standard evaluation approach – we show that PMTs target poverty substantially better than the standard evaluation approach suggests. Across ten countries, their performance is initially better than appears, declines slowly over time, and is indistinguishable from targeting on initial observed consumption – “perfect” targeting – after the first follow-up. Our framework demonstrates that PMTs perform well because 81% of changes in observed poverty reflect transitory shocks and measurement error, while, by design, PMTs target chronic poverty. Consistent with this, over multiple efforts to improve on PMTs – allowing for more sophisticated predictions, and retargeting via updating their models and data – we find limited improvements.

Understanding the implications of observed poverty dynamics for targeting requires a framework for what those dynamics represent. We develop such a framework that decomposes observed poverty into three objects – chronic poverty, transient shocks, and measurement error. The decomposition is inspired by the poverty-dynamics literature (Chaudhuri and Ravallion, 1994; Jalan and Ravallion, 2000; Duclos et al., 2010), but the definition of the components builds on the dynamic-consumption literature, which decomposes consumption into permanent and transitory components (Hall, 1978; Deaton, 1992; Blundell et al., 2008). In our framework, chronic poverty corresponds to low permanent income, and transient poverty corresponds to negative transitory income shocks whose pass-through to consumption depends on household smoothing. In addition, observed poverty further reflects classical measurement error (Chesher and Schluter, 2002; Glewwe, 2012). Empirically, the key distinction between permanent and transitory changes in consumption is that the latter are mean reverting.

We then develop an identification strategy which uses targeting dynamics to separate the three elements of poverty dynamics. Consider a targeting rule which includes those with lowest observed consumption in period 0, which we call observed poverty targeting. This rule is the benchmark against which performance is typically measured, with zero error in period 0 by construction. The increase in error from period 0 to period 1 combines changes in chronic poverty, transitory shocks, and measurement error; while the change from period 1 to period 2 isolates changes in chronic poverty alone. Differencing these two differences identifies the combined transitory and measurement-error, which we then separate with high-frequency data. Namely, we assume that true poverty cannot move discontinuously. An immediate jump in benchmark loss at the instant after baseline therefore identifies measurement error.

To take the framework to the data, we use two complementary sources of panel variation. First, we assemble low-frequency household panels from ten developing countries, drawn from the World Bank’s Living Standard Measurement Surveys (LSMS) and similar living-standards surveys. These panels follow the same households for at least three survey rounds and contain both consumption and the household characteristics used in PMTs, allowing us to trace how observed targeting performance evolves after calibration. They are also typical of the datasets used by governments and international organizations to train and evaluate PMTs (Grosh et al., 2008; Del Ninno and Mills, 2015; Grosh et al., 2022). Second, we use monthly household data from India, where the much shorter spacing between observations helps separate measurement error from genuine short-run consumption shocks. Throughout, we define poverty in terms of relative real per-capita consumption. We measure performance with both mean squared error and with targeting error – the proportion of the population mistargeted. While targeting error is more representative of how targeting is evaluated in practice, it places no weight on the intensive margin of targeting error. It is also highly non-linear, and so ill-suited to our decomposition exercise, which is derived under mean squared error.

Our first result is that the performance of observed poverty targeting – the benchmark against which other methods are measured – deteriorates rapidly at first, then flattens substantially. Across the ten low-frequency country panels, targeting error is zero by construction at calibration, but rises to 25.1 percent by the first follow-up, on average 2.4 years later, and to 26.8 percent by the second follow-up when targeting the poorest 30% of households. The monthly India data show that part of the deterioration happens sharply just after calibration: bottom-30 targeting error jumps to 11.7 percent after one month and rises to 23.4 percent by month 12.

Mapping this to our framework, we find that persistent changes in living standards account for only 19% of the initial observed movement in poverty, while transitory shocks and measurement error account for 81%. This result is consistent with evidence from the poverty-dynamics literature that a large share of observed poverty is transient rather than chronic (Jalan and Ravallion, 1998, 2000; Duclos et al., 2010; Dang and Dabalén, 2019), and with evidence that measurement error can inflate observed mobility and poverty transitions (Glewwe, 2012; Lee et al., 2017). Our contribution is to identify these components from the evolution of targeting loss after calibration. This differs from the main approaches in the poverty-dynamics literature, which decompose panel poverty into chronic poverty at household time-mean consumption and transitory poverty around that mean, infer poverty transitions from repeated cross-sections using synthetic panels, or correct observed transition matrices for measurement error.

The result has three implications for measuring targeting performance. First, measurement error means that standard performance metrics underestimate baseline targeting performance. Second, changes in true poverty are substantial, leading standard performance metrics to overestimate subsequent performance, and bounding how well any static method can target poverty over time. Third, nevertheless, the relatively small changes in chronic poverty suggest that static methods can perform consistently over time, especially if they target chronic poverty initially, in which case initial performance will give a good indication of subsequent performance.

We next apply the framework to re-evaluate the performance of Proxy Means Tests (PMTs),

a targeting method used to decide the eligibility of hundreds of millions of households for welfare programs globally, and among the most criticized under the standard evaluation approach. A PMT is a statistical score built from a household survey: consumption is regressed on characteristics that are cheaper and easier to observe than consumption itself, such as household composition, education, housing quality, utilities, assets, and location (Grosh and Baker, 1995; Coady et al., 2004; Brown et al., 2018). The estimated formula is then applied to the same characteristics collected from applicant households or from a social registry, households are ranked by predicted welfare, and eligibility is assigned below a program cutoff. Like other targeting methods, PMTs have been criticized for large inclusion and exclusion errors (Kidd and Wylde, 2011; Devereux et al., 2017; Brown et al., 2018), even though comparative reviews and field experiments find no consistently superior alternative and often show PMTs performing at least as well as common substitutes (Coady et al., 2004; Alatas et al., 2012; Premand and Schnitzer, 2018).

Consistent with the existing literature, PMTs demonstrate substantial baseline targeting error; however, this is an incomplete picture of their true performance. Their targeting error increases only slowly, and consistently, over time: average PMT classification error is 21.9 percent at calibration, 24.9 percent at the first follow-up, and 26.6 percent at the second follow-up. Thus, after just one round, their performance is indistinguishable from that of observed poverty targeting – “perfect” targeting at time zero. Our mean squared error decomposition uses the increase of the observed poverty targeting benchmark to derive that 70% of the initial PMT error is due to components of observed poverty that, in the case of measurement error, no targeting method should target and, in the case of transient poverty, no static method can hope to target. PMTs thus perform much better at targeting *persistent* deprivation than the standard evaluation suggests.

The stability of PMT performance is a consequence of what PMTs are implicitly designed to measure. We show that, when household characteristics determine permanent income but not transitory shocks or measurement error, the PMT prediction function captures the permanent component of consumption. This is consistent with consumption theory and empirical evidence that consumption is more closely tied to permanent resources than to purely transitory income shocks (Deaton, 1992; Jappelli and Pistaferri, 2017; Paxson, 1992). It is also consistent with the original informational logic of PMTs: when income or consumption is costly to observe directly, observable household characteristics provide proxies for welfare (Ravallion and Chao, 1989; Glewwe et al., 1989; Grosh and Baker, 1995). This interpretation also predicts that static refinements to PMTs should have limited scope to improve performance, a prediction we test directly below.

Given our results that much of observed PMT error reflects transitory poverty and measurement error rather than poor prediction of persistent deprivation, richer static prediction rules should have limited room to improve targeting. We test this hypothesis via three static PMT improvements, in turn helping to explain existing findings in the targeting literature. First, our hypothesis helps explain why previous attempts to improve PMTs with machine learning find modest and context-dependent gains (McBride and Nichols, 2016; Kshirsagar et al., 2017; Areias and Wai-Poi, 2022). In our data, machine learning improves bottom-30 classification error in no country by more than 0.73 percentage points. Second, our hypothesis implies that, once the PMT prediction function targets permanent income, training the PMT on average consumption over multiple periods has limited scope

to improve performance over time. Empirically, the averaged-target PMT performs almost identically to the standard PMT. Third, if PMTs predict permanent income, they cannot anticipate future transitory shocks or permanent innovations. Empirically, training the PMT to predict future poverty with lagged characteristics worsens performance relative to the standard PMT. These results do not imply that improved prediction is never useful; they imply that better static prediction cannot remove the part of observed error that comes from temporary shocks, measurement error, or future changes in persistent living standards.

Retargeting has a different role. Because chronic poverty changes gradually, updating can offset the slow deterioration of a static PMT, but only if the information used to rank households is updated. Annualizing country by country, the standard PMT's bottom-30 classification error rises by 1.03 percentage points per year. Re-estimating the PMT formula while keeping population characteristics stale leaves this deterioration essentially unchanged, removing only 0.01 percentage points per year. Updating household predictors while keeping the baseline formula fixed reduces the increase to 0.22 percentage points per year, removing 0.81 percentage points per year. Fully repeating the targeting exercise removes the increase altogether. This suggests that the fundamental link between household characteristics and chronic poverty changes very slowly, while household characteristics reflect poverty changes over time. Fully repeating the targeting exercise is the most administratively costly option, because it requires recollecting both training data and population predictors. Cheaper updates based on high-frequency information, such as weather, do not appear to substitute for remeasuring the household predictors used to rank households. Governments should therefore retarget when the expected gain in precision is large enough to justify the cost of recollecting the information needed to rank households.

This result connects the paper to a small literature that takes the timing of targeting seriously. Earlier work on dynamic benefit incidence showed that the households reached by social programs need not be the households who were poor when targeting data were collected (van de Walle, 2002; Sumarto et al., 2003). Contemporaneous work studies this problem directly for PMTs and related targeting tools. Hillebrecht et al. (2023) show in Nouna, Burkina Faso, that PMT outperforms community targeting at baseline, but that this advantage disappears 12 and 30 months later. Beuermann et al. (2025) find that in Colombia the exclusion error of a static PMT rises from 30% before the COVID-19 crisis in 2019 to 35% in 2020 during the crises. Aiken et al. (2023) use panel surveys from six low- and middle-income countries and estimate that PMT inclusion and exclusion errors rise by about 1.7 percentage points per year, with data decay roughly three times as important as model decay. Chia and George (2026) examine absolute exclusion and inclusion errors in India and find faster deterioration. Our contribution is to explain these patterns through the decomposition of observed poverty dynamics in ten countries: PMT performance decreases slowly as chronic poverty changes, and updating the PMT removes this increase.

The paper brings insights from models of intertemporal consumption and poverty measurement into the targeting problem, and uses the targeting lens to provide new evidence on poverty dynamics. We find that much of observed consumption mobility reflects transitory shocks, that measurement error also matters, and that less than one fifth of the initial movement reflects changes in permanent living standards. In doing so,

we contribute to an active literature on poverty dynamics (Collins et al., 2009; Merfeld and Morduch, 2026; Armentano et al., 2026). We then use these insights to revisit the targeting problem. Static PMTs perform better than standard evaluations suggest because they target the persistent component of poverty. Their remaining deterioration reflects changes in chronic poverty that can be removed by updating the PMT, but the deterioration is slow.

2. A framework for targeting poverty over time

The implications of poverty dynamics for poverty targeting depend both on their magnitude and what they represent. If households move in and out of observed poverty because of measurement error, standard evaluations will make targeting rules look worse than they are, and there is little reason to retarget on that basis. If observed poverty moves because of transitory shocks, static rules cannot track current need, but they may still target chronic poverty well. If chronic poverty itself changes, then any static rule deteriorates over time. We develop a framework that separates these forces and uses them to define policy-relevant targeting performance.

2.1. Poverty over time

Observed poverty combines persistent deprivation, short-run shocks, and measurement error. We model this mixture with a standard permanent–transitory income process and a simple pass-through parameter that maps transitory income shocks into changes in consumption. This yields the three welfare objects used throughout the paper: chronic poverty, current poverty, and observed poverty.

We begin from a standard permanent–transitory income process (Blundell et al., 2008). Income of household i at time t is

$$y_{it} = p_{it} + u_{it}. \quad (1)$$

Permanent income p_{it} follows a martingale, $p_{it} = p_{i,t-1} + \eta_{it}$, where η_{it} is a mean-zero innovation to permanent income with no serial dependence and a stable distribution over time. As in Santaaulalia-Llopis and Zheng (2018), the transitory component u_{it} captures short-run income fluctuations that are independent over time, have mean zero, and fade within a year. In the settings we study, it is useful to think of permanent shocks as events that shift a household’s longer-run resource path, such as asset loss, sudden bequests, or job displacement, and of transitory shocks as events such as temporary illness, a bad harvest, or short-run price fluctuations.

Households choose consumption based on current resources, expected future resources, and their ability to smooth shocks over time. To allow for different degrees of consumption smoothing in a simple way, we write

$$c_{it} = p_{it} + \alpha u_{it}, \quad \alpha \in [0, 1].$$

The parameter α summarizes how much transitory income shocks pass through to consumption, with shocks to permanent income assumed to pass through fully. When $\alpha = 1$, households are hand-to-mouth and fully absorb transitory shocks in current

consumption. When $\alpha = 0$, households fully smooth transitory shocks and consumption follows permanent income exactly. Intermediate values capture partial smoothing. For simplicity, we assume a common α . Consumption changes according to:

$$\Delta c_{it} = \eta_{it} + \alpha(u_{it} - u_{i,t-1}).$$

This representation yields a simple decomposition of current consumption into a permanent and a transitory part. Because households set the consumption path based on expected future resources, we define permanent consumption as expected next-period consumption,

$$c_{it}^P \equiv \mathbb{E}_t[c_{i,t+1}],$$

and transitory consumption as the deviation from that path,

$$c_{it}^T \equiv c_{it} - c_{it}^P.$$

Because permanent income follows a martingale and the transitory component is mean zero and serially independent, expected next-period consumption equals current permanent income, as shown in [Appendix A.1](#):

$$c_{it}^P = \mathbb{E}_t[c_{i,t+1}] = p_{it},$$

so that transitory consumption is

$$c_{it}^T = \alpha u_{it},$$

and current consumption can be written as

$$c_{it} = c_{it}^P + c_{it}^T. \tag{2}$$

where, empirically, the difference in the dynamics of the chronic and transitory components is that the latter are mean-reverting, while the former are not.

This decomposition lets us distinguish the poverty objects relevant for targeting. Let z_t denote the poverty line in period t . We define chronic poverty as low permanent consumption,

$$c_{it}^P \leq z_t,$$

and current poverty as low current consumption,

$$c_{it} \leq z_t.$$

Finally, researchers and policymakers do not observe c_{it} directly. They observe consumption with error:

$$\tilde{c}_{it} = c_{it} + \omega_{it} = p_{it} + \alpha u_{it} + \omega_{it}, \tag{3}$$

where ω_{it} denotes classical measurement error. Observed poverty therefore differs from both chronic poverty and current poverty. This distinction is central for the rest of the paper: targeting methods may be asked to identify current poverty or chronic poverty, but they should not be judged as if they were meant to target measurement error.

2.2. Targeting over time

Anti-poverty programs aim to reach households who are poor throughout program implementation, not only households who only look poor when the targeting rule is calibrated. Our framework therefore evaluates targeting performance over time and distinguishes poverty from observed poverty.

The planner allocates a scarce anti-poverty program to households in need. The natural benchmark is poverty in the period when benefits are distributed, measured by current consumption c_{it} . Observed consumption $\tilde{c}_{it}$ is only a noisy measure of that object, so observed poverty can differ from poverty because consumption is measured with error. Some programs also have a more persistent objective. A productive asset transfer, such as a graduation-style program, or a program that keeps eligibility fixed for several years, may reasonably aim to reach households facing persistent deprivation.

To evaluate targeting, we use mean squared error as the performance metric, rather than a measure of classification error, because it measures how close a targeting score is to consumption over the full distribution. This avoids making the decomposition depend on a particular poverty line, means that targeting performance depends on both the intensive and extensive margins of targeting errors, and allows us to cleanly decompose the components of error from chronic poverty, transitory poverty, and measurement error. In the empirical analysis, we also report classification error, because the literature often focuses on it, and programs make discrete eligibility decisions.

Let $\hat{c}_{i0}^f$ denote the baseline score generated by a static targeting rule f . The score is used to rank households at the time of targeting and then held fixed. **Standard evaluation** asks how closely the baseline score matches observed consumption at calibration:

$$\mathcal{L}_0^{obs}(f) \equiv \mathbb{E} \left[(\tilde{c}_{i0} - \hat{c}_{i0}^f)^2 \right]. \quad (4)$$

This is the standard statistic computed in a single cross-section, but it is not the object a planner ultimately cares about. It treats observed consumption as the target, even though observed consumption contains measurement error, and it evaluates the rule only at calibration, even though it allocates benefits over time. **Policy-relevant evaluation** instead measures how well the baseline score predicts the program's target in the periods when benefits are distributed:

$$\mathcal{L}_t(f) \equiv \mathbb{E} \left[(c_{it} - \hat{c}_{i0}^f)^2 \right]. \quad (5)$$

We calculate the bias of the standard evaluation criterion by comparing it to this policy-relevant loss. Under the income and measurement-error process above, and for rules that do not target baseline measurement error, let θ_f summarize the extent to which rule f targets transitory poverty at baseline. A rule that targets only persistent deprivation has $\theta_f = 0$, while a rule that also targets baseline transitory consumption has $\theta_f > 0$. Appendix A.2 shows that, for the periods $t > 0$ after calibration, the bias of standard evaluation relative to policy-relevant evaluation is:

$$\mathcal{L}_0^{obs}(f) - \mathcal{L}_t(f; c) = \sigma_\omega^2 - t\sigma_\eta^2 - 2\theta_f\alpha^2\sigma_u^2, \quad (6)$$

where $\sigma_\eta^2 \equiv \text{Var}(\eta_{it})$, $\sigma_u^2 \equiv \text{Var}(u_{it})$, and $\sigma_\omega^2 \equiv \text{Var}(\omega_{i0})$.

This expression gives the central intuition of the framework. Measurement error makes standard evaluation too pessimistic: it raises observed calibration-period loss even though the noise is not part of poverty during implementation. Changes in chronic poverty make standard evaluation too optimistic: a fixed score becomes stale as permanent living standards change. Transitory poverty matters differently. If a static rule targets the baseline transitory shock, that apparent advantage disappears when the shock is replaced by future shocks; this is the negative $2\theta_f\alpha^2\sigma_u^2$ term. If instead the rule targets persistent deprivation, $\theta_f = 0$: transitory shocks still bound how well any static rule can track current poverty, but they do not create a deteriorating loss path.

2.3. An application to Proxy Means Tests

Proxy Means Tests (PMTs) are one of the most widely used static targeting methods in practice. As described in Section 3, PMTs are built from relatively stable household characteristics. As such, the framework predicts that they capture persistent differences in living standards rather than transitory shocks or measurement error. PMTs are therefore a central application of the paper’s main argument: static rules cannot track all movements in current poverty, but they may perform better than standard observed-poverty evaluations suggest when much of that movement is temporary or noisy.

PMTs predict consumption from household characteristics that are easier to collect than consumption itself. In practice, they are typically estimated by regressing observed consumption on observed household characteristics in a training sample and then using the fitted relationship to assign a score in the population:

$$\tilde{c}_i = \beta_0^{PMT} + \beta^{PMT}\Gamma_i + e_i,$$

so that the PMT score used for targeting is

$$\hat{c}_{i0}^{PMT} = \beta_0^{PMT} + \beta^{PMT}\Gamma_{i0}.$$

The key conceptual question is therefore what component of consumption the coefficient β^{PMT} actually captures.

Suppose permanent income is a function of observed household characteristics Γ_i and unobserved characteristics X_i , so that $p_i = g(\Gamma_i, X_i)$. We write the linear projection of permanent income on observed household characteristics as

$$p_i = \beta_0 + \beta\Gamma_i + \varepsilon_i,$$

where $\text{Cov}(\Gamma_i, \varepsilon_i) = 0$. We also assume that the transitory component u_i is the residual short-run shock after persistent household characteristics have been accounted for, so $\text{Cov}(\Gamma_i, u_i) = 0$. This assumption is useful to simplify the derivation below and is consistent with the permanent–transitory income framework, where transitory shocks are temporary innovations rather than predictable household types (Deaton, 1992; Jappelli and Pistaferri, 2017). If observed characteristics predict systematic exposure to recurrent shocks, the framework treats that predictable exposure as part of the persistent component, rather than as the realized transitory shock itself.

PMTs target permanent consumption, because the PMT coefficient identifies the linear projection of permanent income on observable characteristics. Note that the population OLS slope is

$$\beta^{PMT} = \frac{\text{Cov}(\tilde{c}_i, \Gamma_i)}{\text{Var}(\Gamma_i)}.$$

Using $\tilde{c}_i = p_i + \alpha u_i + \omega_i$, we obtain

$$\begin{aligned} \text{Cov}(\tilde{c}_i, \Gamma_i) &= \text{Cov}(p_i + \alpha u_i + \omega_i, \Gamma_i) \\ &= \text{Cov}(p_i, \Gamma_i) + \alpha \text{Cov}(u_i, \Gamma_i) + \text{Cov}(\omega_i, \Gamma_i). \end{aligned}$$

By assumption, transitory shocks and measurement error are orthogonal to observed characteristics, so $\text{Cov}(u_i, \Gamma_i) = 0$ and $\text{Cov}(\omega_i, \Gamma_i) = 0$. It follows that

$$\text{Cov}(\tilde{c}_i, \Gamma_i) = \text{Cov}(p_i, \Gamma_i).$$

Using the projection of permanent income,

$$\begin{aligned} \text{Cov}(p_i, \Gamma_i) &= \text{Cov}(\beta_0 + \beta \Gamma_i + \varepsilon_i, \Gamma_i) \\ &= \beta \text{Var}(\Gamma_i), \end{aligned}$$

and therefore

$$\beta^{PMT} = \frac{\text{Cov}(\tilde{c}_i, \Gamma_i)}{\text{Var}(\Gamma_i)} = \frac{\beta \text{Var}(\Gamma_i)}{\text{Var}(\Gamma_i)} = \beta.$$

As a result, PMTs target chronic consumption rather than transitory fluctuations, and do not target measurement error. The PMT regression residual confirms this result. It shows which components of observed consumption PMTs do not predict, and can be written as

$$e_i = \varepsilon_i + \alpha u_i + \omega_i,$$

as shown formally in Appendix A.3. PMTs therefore miss three components: permanent heterogeneity not captured by observed characteristics (ε_i), transitory consumption fluctuations (αu_i), and measurement error (ω_i).

This characterization also yields a simple expression for PMT performance over time. Evaluated against current consumption, PMT error has three components: a part that rises with changes in permanent consumption after targeting, a part that reflects the PMT's inability to track transitory fluctuations, and a part that reflects permanent heterogeneity not captured by observed characteristics. Appendix A.4 shows that

$$\mathcal{L}_t(\hat{c}_{i0}^{PMT}; c) = t\sigma_\eta^2 + \alpha^2\sigma_u^2 + \sigma_\varepsilon^2,$$

where $\sigma_\varepsilon^2 \equiv \text{Var}(\varepsilon_i)$. The term $t\sigma_\eta^2$ captures changes in permanent consumption after targeting and therefore grows proportionally with time. By contrast, $\alpha^2\sigma_u^2$ and σ_ε^2 are constant over time. $\alpha^2\sigma_u^2$ reflects the inability to track transitory consumption fluctuations that is common to any static targeting rule, while σ_ε^2 reflects baseline mistargeting of the persistent component of consumption.

2.4. Identifying the components of targeting dynamics

Standard targeting evaluations take the baseline observed-poverty ranking as the benchmark. We instead use the dynamics of that benchmark - what happens over time when we target those who appear poorest initially - as an identifying device. Because the benchmark ranks households by observed consumption at calibration, $\tilde{c}_{i0}$, its observed loss is mechanically zero at baseline. Its first post-baseline deterioration captures all forces that make observed poverty move: changes in chronic poverty, transitory shocks, and measurement error. Later deterioration isolates changes in chronic poverty, because the baseline transitory shock and measurement error have already been replaced. The difference therefore identifies the part of observed poverty that any static method cannot persistently target: transitory poverty and measurement error. We then use high-frequency data to separate measurement error from transitory poverty. These objects allow us to recover how well static rules target poverty over time and, in particular, how well they target the chronic component that static methods can hope to track.

For any static targeting rule f , panel data reveal the observed loss path

$$\mathcal{L}_t^{obs}(f) \equiv \mathbb{E}[(\tilde{c}_{it} - \hat{c}_{i0}^f)^2].$$

The policy-relevant loss path depends on what the program means by poverty. If the target is poverty in the evaluation period, observed loss differs from true loss because observed consumption contains measurement error, so $\mathcal{L}_t^{obs}(f) - \mathcal{L}_t(f; c) = \sigma_\omega^2$ for horizons after calibration. But a static targeting rule cannot persistently track the transitory component of poverty: once the baseline score is fixed, future transitory shocks are newly realized. The object that a static method can hope to target over time is therefore chronic poverty. Relative to that object, observed loss also contains the transitory component, so the relevant wedge is

$$W \equiv \alpha^2 \sigma_u^2 + \sigma_\omega^2.$$

Thus, $\mathcal{L}_t^{obs}(f) - \mathcal{L}_t(f; c^P) = W$. These expressions follow from the loss decompositions above. The empirical task is to identify W , because it is the part of observed loss that reflects transitory poverty and measurement error rather than chronic mistargeting, and, when possible, to split it into its two components.

Low-frequency panels identify the combined wedge W . For the baseline observed-poverty benchmark, Appendix A.5 shows that observed loss follows

$$\mathcal{L}_t^{obs}(\tilde{c}_{i0}) = \begin{cases} 0, & \text{for } t = 0, \\ t\sigma_\eta^2 + 2W, & \text{for } t > 0. \end{cases}$$

The first post-baseline deterioration combines changes in chronic poverty, transitory shocks, and measurement error. Later deterioration isolates changes in chronic poverty, because the baseline transitory shock and baseline measurement error have already been replaced. With unequal survey spacing, let t_1 denote the first follow-up year and $t_2 > t_1$ a later follow-up year. Then

$$\sigma_\eta^2 = \frac{\mathcal{L}_{t_2}^{obs}(\tilde{c}_{i0}) - \mathcal{L}_{t_1}^{obs}(\tilde{c}_{i0})}{t_2 - t_1}, \quad (7)$$

and

$$W = \alpha^2 \sigma_u^2 + \sigma_\omega^2 = \frac{1}{2} [\mathcal{L}_{t_1}^{obs}(\tilde{c}_{i0}) - t_1 \sigma_\eta^2]. \quad (8)$$

The equal-spacing case is the special case in which $t_1 = 1$ and $t_2 = 2$. Appendix A.6 gives the derivation.

Once W is identified, observed PMT loss can be translated into chronic-poverty loss. Subsection 2.3 shows that PMTs target chronic consumption rather than transitory shocks or measurement error. Therefore, at calibration and at any later horizon,

$$\mathcal{L}_t(\hat{c}_{i0}^{PMT}; c^P) = \mathcal{L}_t^{obs}(\hat{c}_{i0}^{PMT}) - W = t\sigma_\eta^2 + \sigma_\varepsilon^2. \quad (9)$$

In words, subtracting W strips out the part of observed PMT error that comes from evaluating a chronic-poverty tool against noisy observed consumption. What remains is baseline chronic mistargeting, σ_ε^2 , plus the gradual deterioration from changes in chronic poverty, $t\sigma_\eta^2$.

High-frequency data separate measurement error from transitory poverty. Low-frequency panels identify W , but not its two components. The additional identifying idea is that true consumption cannot jump discontinuously at the instant after calibration, while measurement error can be redrawn immediately. Let

$$m(t) \equiv \mathbb{E}[(\tilde{c}_{it} - \tilde{c}_{i0})^2]$$

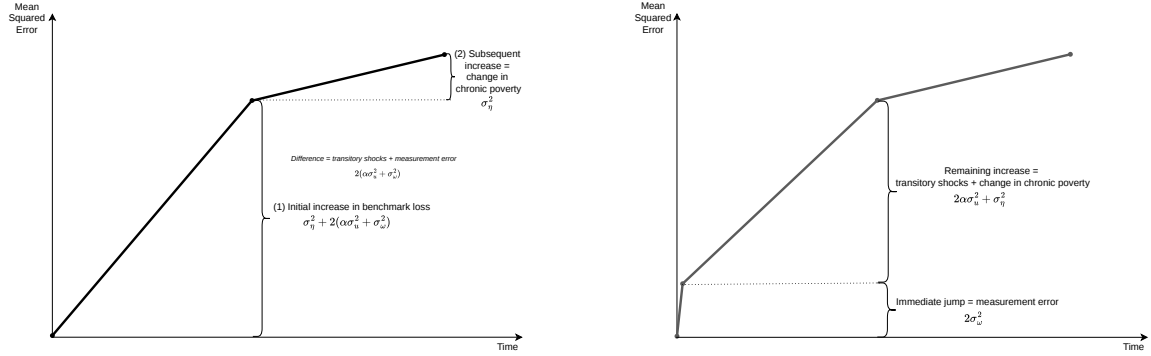
denote the observed loss of the baseline benchmark at elapsed time t after calibration. Appendix A.7 shows that

$$\Delta_0 \equiv \lim_{t \downarrow 0} [m(t) - m(0)] = 2\sigma_\omega^2. \quad (10)$$

This identifies $\sigma_\omega^2 = \Delta_0/2$. The transitory component is then the residual, $\alpha^2 \sigma_u^2 = W - \sigma_\omega^2$, and current-poverty loss can be recovered as $\mathcal{L}_t(f; c) = \mathcal{L}_t^{obs}(f) - \sigma_\omega^2$.

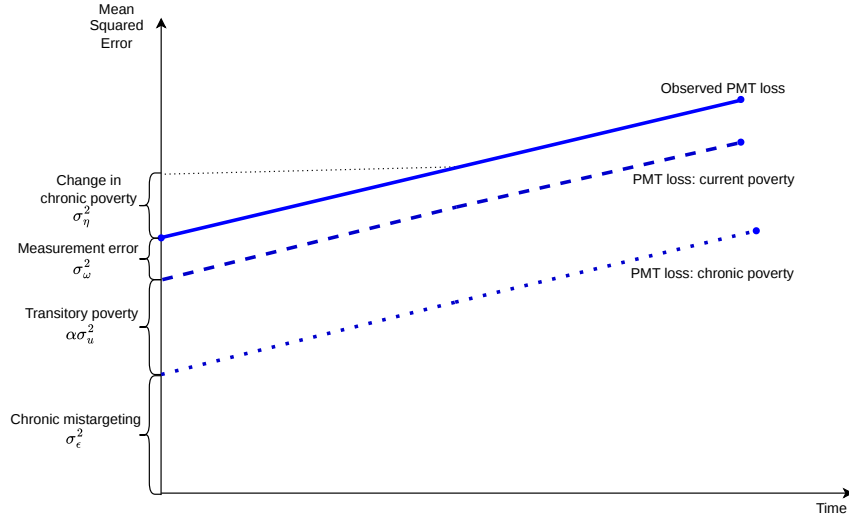
Figure 2 summarizes the identification logic. Panel (a) shows how low-frequency benchmark dynamics identify chronic changes and the combined wedge W . Panel (b) shows how high-frequency data identify measurement error from the immediate jump in observed benchmark loss. Panel (c) applies the pieces to PMTs: observed PMT loss combines chronic mistargeting, transitory poverty, measurement error, and gradual changes in chronic poverty.

Figure 2 — Stylized decomposition of targeting performance over time



(a) Low-frequency identification

(b) High-frequency identification



(c) Implications for PMT performance

Notes: The figure summarizes the identification logic in Section 2.4. Panel (a) shows that low-frequency data identify the combined transitory and measurement-error wedge from the evolution of the baseline observed-poverty benchmark. Panel (b) shows that high-frequency data identify measurement error from the immediate jump in benchmark loss after calibration. Panel (c) combines these pieces to decompose observed PMT loss into chronic mistargeting, inability to track transitory poverty, measurement error, and changes in chronic poverty over time. All panels are drawn in terms of mean squared error.

3. Data and empirical implementation

The empirical design combines two sources of panel variation. Ten low-frequency household panels show how static targeting rules perform as baseline information ages in the survey settings used to calibrate PMTs. A monthly panel from India adds the short-horizon variation needed to separate measurement error from transitory poverty. Together, these data let us estimate observed targeting loss paths and translate them into the chronic, transitory, and measurement-error components derived in Section 2.

3.1. Low- and high-frequency panel data

The low-frequency sample contains ten pre-Covid household panels with repeated consumption and PMT-style predictors; the high-frequency sample contains monthly food-consumption data from rural India.

The low-frequency panels summarized in Table 1 are selected to match the targeting problem faced by anti-poverty programs. Each panel contains a consumption-based welfare measure, PMT-style household characteristics, at least three pre-Covid survey rounds, and a sample large enough to estimate a PMT in one subsample and evaluate it in another. The panels represent national populations in most countries, rural Bangladesh, rural and small-town Ethiopia, and the regional rural population covered by the Thailand panel. Following the same households over time lets us hold a baseline targeting rule fixed and ask how its performance evolves after calibration. Observing household characteristics in later rounds also supports the updating exercises in Section 5.

We construct real per-capita consumption in each country and use its logarithm for mean-squared-error analysis. This follows standard poverty-measurement practice in low- and middle-income countries, where consumption is preferred to income because it is closer to material living standards and easier to measure in informal, self-employed, agricultural, and in-kind settings (Deaton, 1997; Grosh and Glewwe, 2000; Ravallion, 2016). The welfare aggregates are designed to capture broad household consumption: food at home and away from home, regular non-food spending, own-produced consumption, gifts and in-kind receipts, housing services, and education or health spending where these are included by the survey. Consistent with standard aggregate construction, durable purchases and owner-occupied housing are treated differently from ordinary current spending because the relevant welfare object is the service flow rather than the purchase price (Deaton and Zaidi, 2002). We use survey-provided real or price-adjusted aggregates where available and otherwise construct comparable annual aggregates from the survey modules. PMT predictors are observable household characteristics that can plausibly be collected or verified in a targeting exercise. We harmonize them into six families: geography, household composition, head characteristics, housing, consumer assets, and productive assets. Appendix B reports the panel construction, consumption choices, and predictor coverage behind Table 1.

The monthly India panel complements the low-frequency cross-country panels by showing what very short horizons reveal about the sources of targeting error. We use the Village Dynamics in South Asia (VDSA) project collected by ICRISAT, which covers 30 villages in 15 districts across five states. The balanced analysis panel contains 616 households observed monthly from July 2010 to June 2015 for food consumption and annually for standard PMT household characteristics. The India panel is local rather than nationally representative, but its monthly frequency lets us observe the immediate post-baseline jump in benchmark loss that identifies measurement error in Section 2.4.

Table 1 — Panel datasets used in the analysis

Country	Years used	Households	Predictors	Coverage
<i>Panel A. Low-frequency panels</i>				
Bangladesh	2011/12, 2015, 2018/19	5,010	21	Rural
Ethiopia	2011/12, 2013/14, 2015/16	2,301	68	Rural/small-town
Ghana	2009/10, 2013/14, 2017/18	3,553	24	National
Malawi	2010/11, 2013, 2016, 2019	1,458	24	National
Nigeria	2010/11, 2012/13, 2015/16	3,720	52	National
Peru	2013, 2014, 2015	6,979	23	National
Tanzania	2008/09, 2010/11, 2012/13	3,055	74	National
Thailand	2007, 2008, 2010, 2013, 2016, 2017	1,269	14	Regional
Uganda	2013/14, 2015/16, 2018/19, 2019/20	1,770	43	National
Vietnam	2014, 2016, 2018	1,806	57	National
<i>Panel B. High-frequency panel</i>				
India	Monthly, 2010–2015	616	53	Local

Notes: Households denotes the balanced analytical panel used in the targeting analysis. All low-frequency panels use standard annual real per-capita consumption; India uses monthly per-capita food consumption. The predictors column counts conceptual PMT predictors: categorical variables generated from one survey question count once, while distinct asset items count separately. Appendix B reports panel construction, consumption construction, and predictor coverage.

3.2. Constructing PMTs, the baseline benchmark, and targeting loss paths

The empirical exercise compares two static scores: a baseline PMT and the baseline observed-poverty benchmark. Both are evaluated out of sample as the same household panels age.

We construct two baseline targeting rules. The first is the standard PMT: in each training sample, we regress baseline log real per-capita consumption on household characteristics using ordinary least squares and use the fitted rule to generate the baseline PMT score $\hat{c}_{i0}^{PMT}$ in the hold-out sample. That score is then held fixed over time. The second is the baseline observed-poverty benchmark, which uses observed baseline consumption itself as the targeting score. This benchmark matters both because it mirrors the standard evaluation logic in the targeting literature and because its observed loss path is the key object used to recover chronic poverty, transitory poverty, and measurement error.

We evaluate these static rules using observed loss over time. The main metric is mean squared error,

$$\mathcal{L}_t^{obs}(f) = \mathbb{E} \left[(\tilde{c}_{it} - \hat{c}_{i0}^f)^2 \right],$$

because it maps directly into the framework in Section 2 and uses information from the full consumption distribution. By contrast, classification error depends on a poverty cutoff and treats all targeting errors equally: excluding a household just below the line and excluding one far below it both count as one exclusion error, and the same is true for

inclusion errors above the line. Mean squared error is therefore better suited to our main decomposition, which separates chronic poverty, transitory poverty, and measurement error across the distribution rather than only at one threshold. We report classification error as a secondary measure because it is standard in the targeting literature and easy to interpret for discrete eligibility decisions. We define relative poverty as the bottom 30 percent of the country-wave distribution so that targeting rules are compared at a common coverage rate.

We estimate these observed loss paths out of sample. In each low-frequency panel, we randomly split households into a training sample used to estimate the PMT and a hold-out sample used to evaluate both the PMT and the baseline benchmark. We then carry the baseline scores forward and compute observed loss at each horizon in the hold-out sample. To reduce sensitivity to any particular split, we repeat this exercise 100 times and average the resulting loss paths. Most figures in the paper report equally weighted averages across country panels, while the appendix reports country-specific results. For country-average low-frequency figures, we construct 95% confidence intervals across country panels after averaging over repeated splits within each country. The high-frequency analysis follows the same logic but reports confidence intervals across repeated splits.

3.3. Empirical implementation of the targeting loss decomposition

The low-frequency panels identify the combined transitory-poverty and measurement-error wedge. The monthly India panel separates measurement error from that wedge.

The first step uses the low-frequency panels. In each country, we estimate the observed loss path of the PMT and of the baseline benchmark at each available horizon. We then apply the low-frequency decomposition from Section 2.4: equation (7) uses post-baseline benchmark deterioration to identify the annual change in chronic poverty, and equation (8) uses the first follow-up loss net of this chronic component to identify the combined wedge of transitory poverty and measurement error. Subtracting that wedge from observed PMT loss yields policy-relevant PMT loss relative to chronic poverty, as in equation (9). In the country panels where the benchmark path contains large wave-specific movements that also appear in the PMT path, we use the PMT-assisted extension described in the results table notes. This keeps the decomposition tied to the same model but avoids forcing a smooth chronic-poverty path on country episodes with large aggregate shocks.

The second step uses the monthly India data. There, we use the shortest observed horizon to approximate the immediate post-baseline jump in benchmark loss, as in equation (10). Under the continuity condition in Section 2.4, this jump identifies measurement error directly. Once measurement error is pinned down, current-poverty loss for any static targeting rule follows from $\mathcal{L}_t(f; c) = \mathcal{L}_t^{obs}(f) - \sigma_\omega^2$, and transitory poverty is recovered as the difference between the low-frequency wedge and the measurement-error component. Figure 2 summarizes these two empirical steps and guides the results in the next section.

4. Results: poverty and targeting performance over time

Poverty is too dynamic for calibration-period performance to be a reliable and comparable measure of subsequent targeting performance. Across the ten low-frequency country

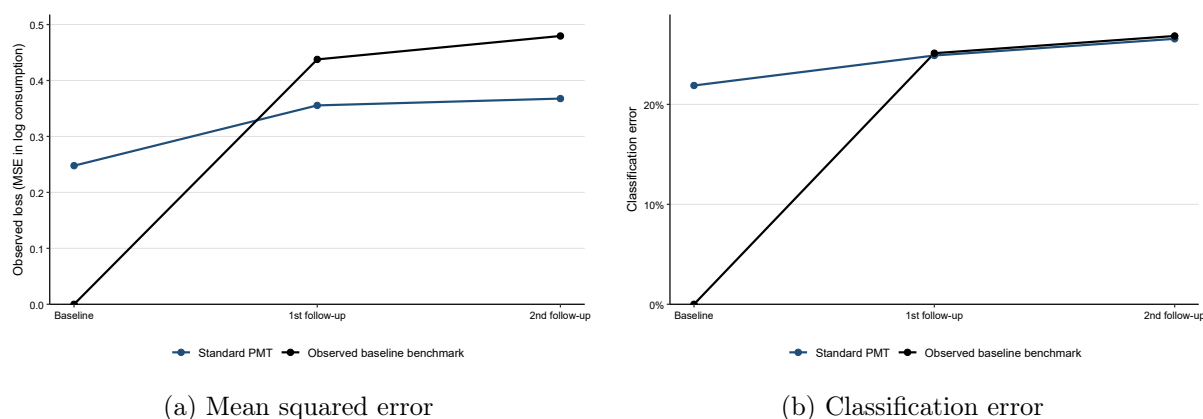
panels, the observed-poverty benchmark has zero error by construction at calibration, but its MSE rises to 0.438 by the first follow-up. Decomposing that first deterioration shows that only 0.098 reflects movement in persistent living standards, while 0.340 reflects transitory poverty and measurement error. This distinction changes the interpretation of PMTs: their average baseline observed MSE is 0.248, but the chronic-poverty mistargeting component is only 0.075 after subtracting the transitory-plus-measurement wedge. PMTs therefore perform substantially better than standard observed-poverty evaluations suggest for the persistent component of poverty that static rules can plausibly target, while no static rule can track current poverty well when temporary shocks are important.

4.1. Decomposing poverty and targeting over time

The observed-poverty benchmark deteriorates sharply after calibration, but most of that deterioration is not movement in chronic poverty. In the average country, transitory poverty and measurement error account for 81 percent of the first-follow-up benchmark loss.

Observed poverty dynamics are large, and this invalidates the standard evaluation approach that judges targeting against observed poverty at calibration. Figure 1 showed large reshuffling of households in the observed consumption distribution. In the targeting framework, the same fact appears as rapid deterioration of the baseline observed-poverty benchmark, shown in Figure 3. Its MSE rises from 0 at calibration to 0.438 at the first follow-up and 0.480 at the second follow-up. In classification terms, the same benchmark's bottom-30 error rises from 0 to 25.1 percent by the first follow-up and 26.8 percent by the second. The same pattern appears country by country in Appendix Figures C.3 and C.4, while Appendix Tables C.1 and C.2 show the underlying absolute and rank mobility. Once observed poverty itself moves this quickly after calibration, evaluating a static targeting rule against observed poverty in the calibration period cannot be a good guide to targeting performance over time.

Figure 3 — Observed targeting performance over time



Notes: The figure reports observed targeting performance over time for the standard PMT (blue) and the baseline observed-poverty benchmark (black), averaged across the ten low-frequency country panels. Panel (a) reports mean squared error in log real per-capita consumption. Panel (b) reports bottom-30 classification error. The benchmark uses observed baseline consumption itself as the targeting score. The figure is based on 100 repeated train-test splits. Country-specific figures are reported in Appendix C.2.

The central empirical question is what drives those observed poverty dynamics, because that determines what static targeting methods can realistically achieve. If observed poverty dynamics mainly reflect transitory shocks and measurement error, static methods may still perform much better at targeting persistent deprivation than standard evaluation suggests. If they instead reflect rapid changes in chronic poverty, then any static method will deteriorate quickly. To separate these cases, we use the baseline observed-poverty benchmark. Its loss is

$$\mathcal{L}_t^{obs}(\tilde{c}_{i0}) = \mathbb{E}[(\tilde{c}_{it} - \tilde{c}_{i0})^2],$$

which is the mean squared change in observed consumption from baseline and therefore a structured summary of how observed poverty evolves over time. Section 2.4 shows how the model decomposes that observed evolution into chronic poverty, transitory poverty, and measurement error; Section 3 describes the empirical implementation. We now bring that decomposition to the data.

Little of the observed short-run churn is churn in chronic poverty. Section 2.4 shows that the first increase in benchmark loss combines movement in chronic poverty with the replacement of transitory shocks and measurement error, while later movements identify the persistent component. Applying that decomposition country by country, Appendix Table C.3 shows that the first-follow-up benchmark loss averages 0.438. Of this loss, 0.098 is the estimated persistent component and 0.340 is the combined transitory-poverty and measurement-error component. In country-level shares, the residual component explains 81 percent of initial observed movement. Ghana and Malawi require the PMT-assisted extension described in Section 3 because both the observed benchmark and the PMT contain large wave-specific movements; Appendix Figure C.6 shows that the main targeting pattern is not driven by those two panels. Through the targeting lens, Figure 1 therefore acquires a clear interpretation: the large reshuffling of households in observed consumption over short horizons mostly does not represent households abruptly switching chronic-poverty status. It reflects temporary movements and survey noise layered on top of slower-moving changes in persistent deprivation.

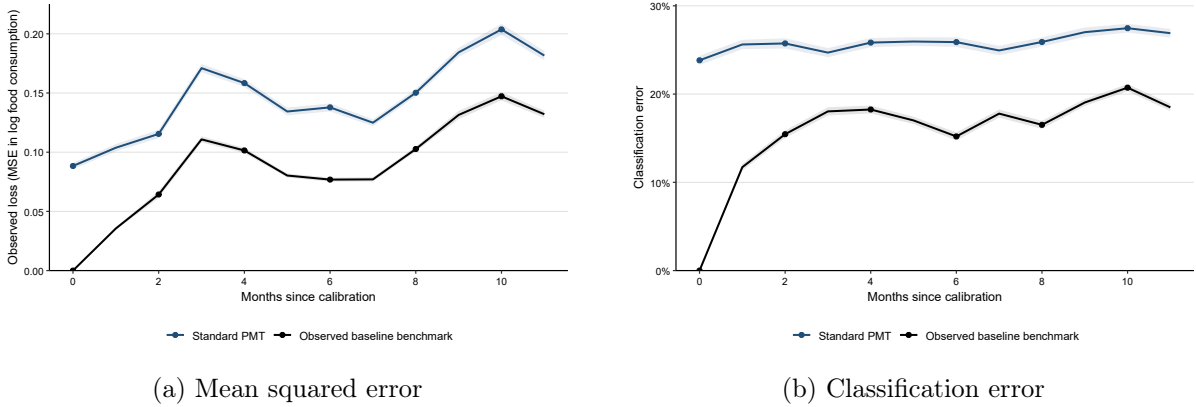
The monthly India data show that measurement error matters, but it is too small to explain most short-run poverty churn; the remainder reflects genuine transitory poverty. For programs such as food aid or temporary relief, the relevant question is not only how much observed poverty churn reflects chronic poverty, but also how much of the remaining short-run wedge reflects transitory poverty rather than replacement of measurement error. To address that question, we use the monthly India data. Because these data track monthly food consumption in a smaller local sample, their levels are not directly comparable to the annual total-consumption panels. Their value is instead that they let us apply the short-horizon logic summarized in Panel B of Figure 2 and establish orders of magnitude. Figure 4 reports the monthly path over the first year. Benchmark MSE rises from 0 at baseline to 0.036 after one month. Under the continuity condition in Section 2.4, this immediate jump identifies the contribution of measurement error to observed short-run poverty dynamics:

$$\widehat{2\sigma_\omega^2} = \widehat{\mathcal{L}}_1^{obs}(\tilde{c}_{i0}) - \widehat{\mathcal{L}}_0^{obs}(\tilde{c}_{i0}) \approx 0.036.$$

Measurement error is therefore non-negligible, and Panel B shows that it immediately produces an 11.7 percentage-point classification error for the observed benchmark. But

measurement error cannot explain the full benchmark path. Benchmark MSE continues to rise to 0.132 by month 12, indicating an important role for transitory consumption changes, including seasonal movement in food consumption. The monthly data therefore imply that genuine transitory poverty is quantitatively important. The implication for targeting is direct: static methods may still target persistent deprivation well, but they cannot be expected to track current poverty well when temporary shocks account for a large share of short-run poverty dynamics.

Figure 4 — Observed benchmark deterioration at monthly frequency



Notes: The figure reports monthly targeting performance over the first year of the India panel for the standard PMT (blue) and the baseline observed-poverty benchmark (black). Panel (a) reports mean squared error in log food consumption. Panel (b) reports bottom-30 classification error. The figure is based on 100 repeated train-test splits. The benchmark's initial jump in panel (a) is the empirical object used to identify measurement error at short horizons, while the continued movement in both panels shows that short-run poverty churn is not explained by measurement error alone. Appendix Figure C.8 reports the full 60-month path.

The decomposition yields three lessons for targeting. First, measurement error mechanically raises observed loss, and the monthly India data show that this bias is not negligible: the immediate measurement-error jump in benchmark MSE is 0.036. For a static method that does not target measurement error at calibration, observed loss therefore overstates true loss in every period. Second, transitory poverty is quantitatively important: in India, the benchmark MSE more than triples between month 1 and month 12, from 0.036 to 0.132. No static rule can track that part of current poverty after calibration. Third, the low-frequency panels are much more encouraging about the persistent component of poverty. Across countries, only 0.098 of the 0.438 first-follow-up benchmark loss reflects persistent movement, while 0.340 reflects transitory poverty and measurement error. Methods that capture persistent deprivation can therefore perform much better over time than the standard observed-poverty evaluation framework suggests. We now apply these insights to PMTs, one of the most important targeting methods in practice.

4.2. PMT performance over time

Observed PMT error substantially overstates chronic-poverty mistargeting. Average baseline PMT MSE is 0.248 against observed consumption, but only 0.075 against the

persistent component after subtracting the transitory-poverty and measurement-error wedge.

The decomposition above changes how the PMT line in Panel (a) of Figure 3 should be read. Section 2.3 showed that PMTs predict the permanent component of consumption encoded in household characteristics. Observed PMT loss therefore mixes four distinct forces: baseline mistargeting of persistent deprivation, subsequent changes in that persistent component, inability to track transitory poverty, and measurement error in observed consumption. The relevant question is not whether observed PMT error is high, but how much of that error reflects a failure to identify the component of poverty that a static rule can reasonably target. Panel C of Figure 2 provides the map: the blue line can be decomposed into chronic mistargeting, transitory poverty, measurement error, and gradual changes in chronic poverty over time.

PMTs perform much better for the persistent component of poverty than their observed error suggests. Using the low-frequency decomposition from the previous subsection, policy-relevant PMT loss relative to chronic poverty is

$$\begin{aligned}\mathcal{L}_t(\widehat{\hat{c}}_{i0}^{PMT}; c^P) &= \mathcal{L}_t^{obs}(\widehat{\hat{c}}_{i0}^{PMT}) - \alpha^2 \widehat{\sigma_u^2} + \sigma_\omega^2 \\ &\approx \mathcal{L}_t^{obs}(\widehat{\hat{c}}_{i0}^{PMT}) - 0.173.\end{aligned}$$

At baseline, observed PMT loss is about 0.25, so the implied chronic-poverty loss is only about 0.075. In other words, roughly 70 percent of the PMT’s observed baseline error reflects transitory poverty and measurement error rather than imperfect targeting of chronic poverty. Appendix Table C.4 reports the same calculation by country. The correction is large in nearly all panels: the wedge accounts for more than half of baseline observed PMT loss in eight of the ten countries, and for 43–44 percent even in Peru and Vietnam, where it is smallest. The same pattern appears in classification terms in Panel (b) of Figure 3. PMT classification error for the bottom 30 percent is 21.9 percent at baseline, 24.9 percent at the first follow-up, and 26.6 percent at the second follow-up. Appendix Table C.5 shows that, by the final observed wave, average classification error rises from 22.0 to 27.0 percent, an increase of 0.91 percentage points per year. Appendix Figure C.9 shows that the qualitative pattern remains when households in a 20-percentile band around the poverty line are excluded, so the result is not only about marginal errors. Appendix Figure C.10 reaches the same conclusion from a different angle: after baseline shocks and measurement error are replaced, the benchmark and the PMT target similar shares of households across the observed consumption distribution.

Chronic-poverty PMT performance also deteriorates only slowly over time. Combining the baseline chronic-poverty loss with the annual change in chronic poverty reported in Appendix Table C.4 yields

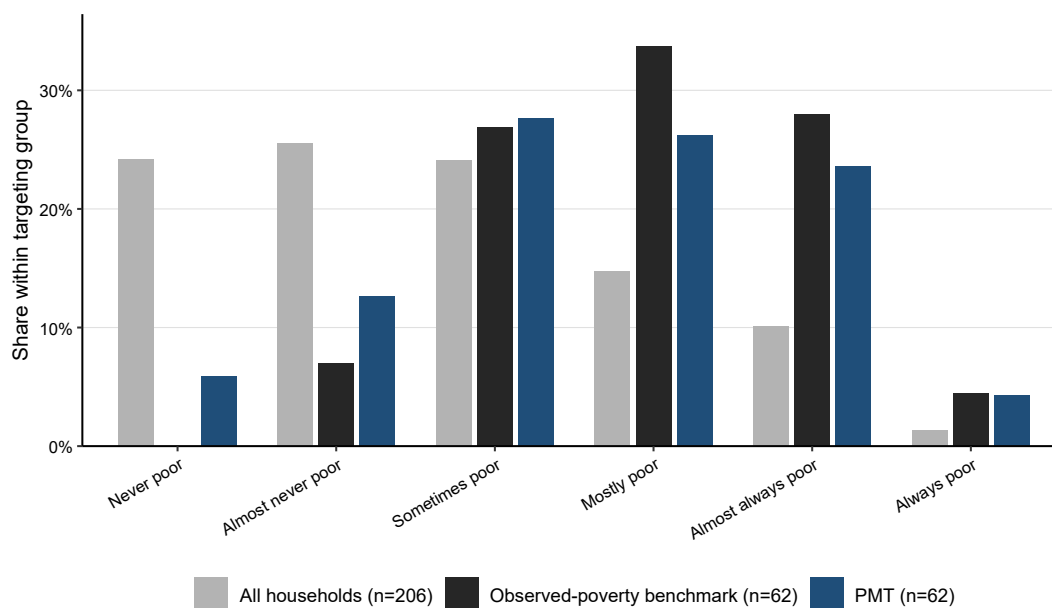
$$\mathcal{L}_t(\widehat{\hat{c}}_{i0}^{PMT}; c^P) \approx 0.075 + 0.027t.$$

This is the model-based counterpart of a simple empirical fact already visible in the blue line: PMT loss rises much more slowly than the benchmark’s initial jump. Observed PMT MSE increases by 0.020 per year on average, close to the estimated 0.027 annual movement in chronic poverty. That is exactly what the framework predicts if PMT

deterioration mainly reflects gradual movement in persistent living standards, while the benchmark’s sharp initial deterioration is driven by transitory poverty and measurement error. The central implication is that the standard evaluation framework makes PMTs look much worse than they are for the part of poverty a static method can target. PMTs do not perfectly identify persistent deprivation, but they identify it much better, and more durably, than a comparison to observed poverty at calibration would suggest.

A descriptive look at poverty duration reinforces the interpretation that PMTs identify persistently poor households while excluding households that are persistently non-poor. Figure 5 groups households in the monthly India panel by the number of months they spend in poverty, according to observed food consumption and a 30% relative poverty line over the full five-year period, and shows the poverty-duration distribution within each targeting group. If a targeting rule captured no persistent poverty signal, its selected households would look like the full sample. Instead, the PMT group is concentrated among households with long poverty exposure and contains relatively few households that are never poor. The baseline benchmark is even more concentrated at the extremes, as expected given its direct use of observed consumption at calibration, but the PMT already captures much of the persistent poverty structure in the data.

Figure 5 — Poverty duration by targeting group



Notes: The figure reports the distribution of poverty duration within each targeting group in the India panel. Poverty duration is the number of months in poverty over the 60-month panel, using monthly food consumption and a bottom-30 relative poverty line. Bars report shares within each group, not counts. The PMT and observed-poverty benchmark groups are averaged across 100 repeated train-test splits.

The picture is less favorable when the policy objective is current poverty rather than persistent deprivation. A PMT can identify persistent deprivation reasonably well, but current poverty also includes transitory deviations from the consumption path. Because those temporary shocks are part of the target, a static PMT will miss households whose

need arises after calibration even if it targets chronic poverty well. The monthly India evidence in Figure 4 shows that this transitory component is quantitatively important: after the immediate measurement-error jump, benchmark MSE continues to rise by another 0.096 by month 12. Static PMTs therefore cannot be expected to track current poverty well over time. This is a real limitation, but it is not a special failure of PMTs; it is a limitation of any static targeting method in an environment where temporary shocks are economically important. Standard evaluation against observed poverty at calibration therefore confounds two distinct critiques: genuine mistargeting of persistent deprivation and the inability of any static rule to track current need.

Taken together, the results suggest that PMTs are substantially better than standard evaluations imply at targeting the part of poverty that an infrequently updated static rule can plausibly target. In the average country, subtracting the transitory-poverty and measurement-error wedge reduces baseline PMT loss from 0.248 to 0.075, and observed bottom-30 classification error rises by only 0.91 percentage points per year. The familiar critique that PMTs perform poorly is therefore overstated when the objective is to reach persistently poor households. But the results also imply a clear limit: programs that aim to track current need require more dynamic information than any static PMT can provide. The next section turns from interpretation to design and asks whether alternative static or dynamic rules can materially improve on the standard PMT.

5. Implications for targeting design

The results above show that PMTs are best understood as tools for targeting chronic poverty rather than current poverty. This section confirms this result by showing theoretically and empirically that variants of the PMT keep targeting chronic poverty. Variants of the static PMT achieve limited reductions of the chronic poverty loss, showing that the standard PMT already performs well, and are unable to anticipate changes in chronic poverty, consistent with our consumption model. Dynamic PMTs, which update the beneficiary list, can remove the mechanical increase in targeting error caused by stale targeting information, but do so most effectively only when the whole targeting exercise is repeated, which is expensive.

5.1. Static refinements to PMTs

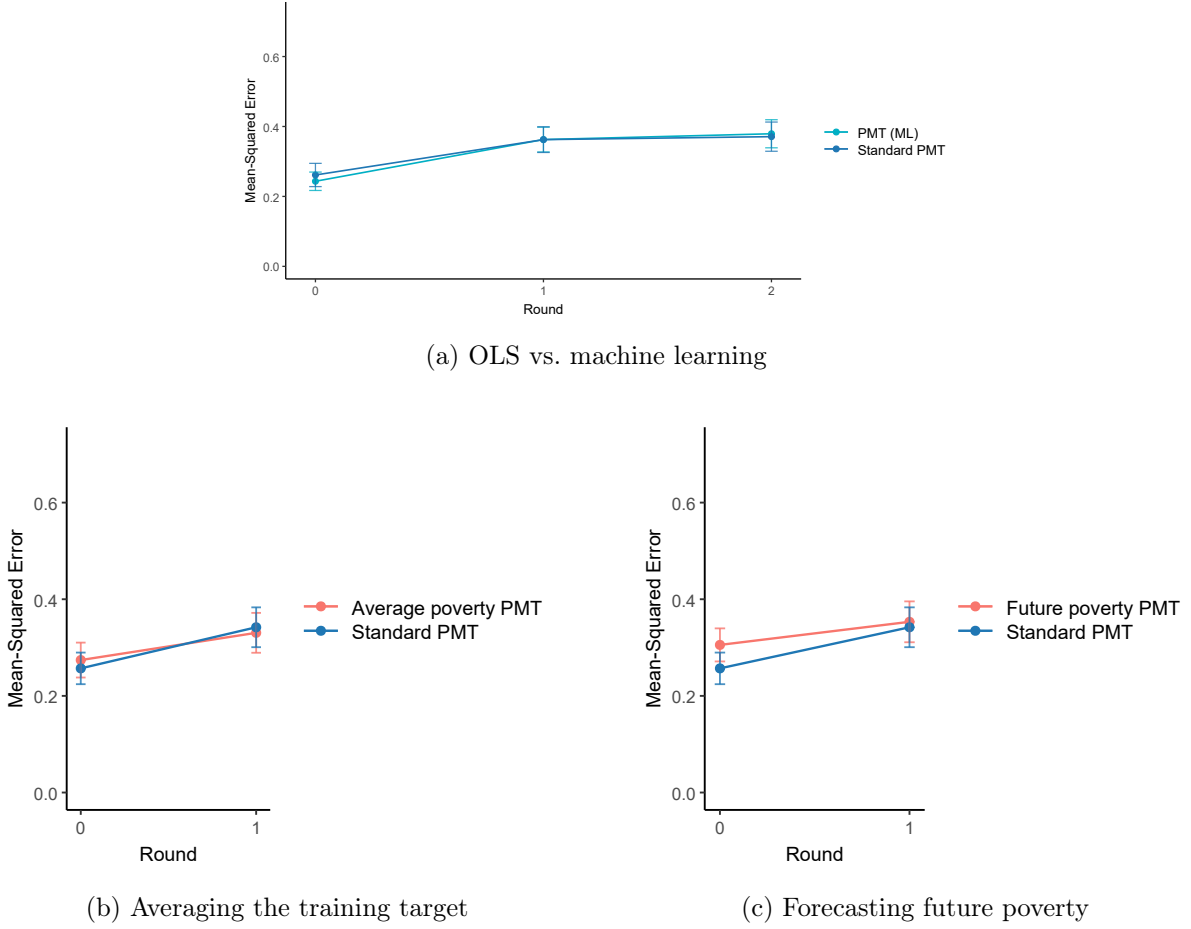
Refining PMTs via machine learning prediction, a target of average consumption across two periods, and predicting future consumption yields limited or no improvements over the standard PMT. These results are in line with our model and confirm that the standard PMT does a good job at capturing baseline chronic poverty.

The extension of the model shows that the theoretical scope for improving PMTs through static refinements is narrow. Appendix D.1 first shows that any static PMT, no matter how sophisticated its prediction method, targets the chronic component of consumption through the signal contained in household characteristics rather than transitory shocks or measurement error, which is precisely the relevant object for a static method. It then shows that static refinements can improve performance only by reducing baseline chronic-poverty prediction error, and only to the extent that the standard PMT leaves predictable chronic-poverty signal unused. In that sense, more flexible prediction methods

such as machine learning may help at the margin, but they cannot change the part of targeting error that comes from transitory poverty or from subsequent changes in chronic poverty after targeting. The same logic applies to redefining the training target: averaging observed consumption over several periods may reduce noise in the dependent variable, but it does not change the underlying chronic-poverty signal that a static PMT can capture in the population. Finally, a PMT trained to predict future consumption cannot anticipate future transitory shocks or permanent innovations that arrive after targeting; at best, it predicts the same chronic component using older information. The empirical exercises below test the theoretical predictions.

Machine learning yields only limited improvements over the standard PMT, confirming that the main remaining problem is not misspecification of the prediction rule but poverty dynamics after targeting. Proposition 5 in Appendix D.1 shows that a machine learning PMT can improve performance if it predicts the permanent component of consumption better at baseline; like any version of the PMT, it does not target transitory poverty or measurement error, nor can it anticipate the error increase caused by subsequent changes in chronic poverty. Figure 6a is consistent with exactly that prediction. The machine learning PMT performs similarly to the standard OLS PMT. The interpretation is not that machine learning is unhelpful in general, but that the standard PMT already extracts much of the chronic-poverty signal available in baseline characteristics.

Figure 6 — Static refinements to PMTs



Notes: The figure compares the observed performance over time of the standard static PMT to alternative static refinements. Panel (a) compares the standard OLS-based PMT to a machine learning PMT using the same baseline predictors. Panel (b) compares the standard PMT to an averaged-target PMT trained on first-wave characteristics with average consumption over the first two waves as the target. Panel (c) compares the standard PMT to a future-poverty PMT trained on first-wave characteristics to predict second-wave consumption. Panels (b) and (c) therefore use the first wave as a pre-baseline period, so the ten-country figure displays the one common follow-up available in all panels. The outcome is observed mean squared error, averaged across the ten low-frequency country panels and 100 repeated train-test splits. Error bars show 1.96 times the average standard deviation across repeated splits within countries.

Averaging the training target yields no improvement, confirming the model’s prediction that the standard PMT already identifies chronic poverty. A natural way to make the PMT look more like a chronic-poverty tool is to train it on average consumption over several pre-periods rather than on one noisy consumption observation. This can reduce transitory fluctuations and measurement error in the dependent variable used for training. But Proposition 6 in Appendix D.1 shows that, in the model, this refinement does not change the structural signal the PMT captures in the population: a PMT trained on an averaged target still identifies the same component of permanent consumption encoded in observed characteristics as the standard PMT. The main potential gain is therefore finite-sample stabilization rather than a structural reduction in chronic-poverty mistargeting. Figure 6b

is consistent with exactly that prediction. The PMT trained on an averaged target performs almost identically to the standard PMT, indicating that the standard PMT already captures most of the chronic-poverty signal available in baseline characteristics.

Training a PMT to predict future consumption also yields no improvement, confirming that static methods cannot anticipate poverty dynamics with baseline information alone. This is the most forward-looking static refinement: instead of predicting consumption at calibration, it trains the PMT to predict consumption in a future period. But Proposition 7 in Appendix D.1 shows that, in the model, a forecasting PMT still identifies only the permanent component encoded in lagged observables. It cannot anticipate future transitory shocks, and it cannot predict permanent innovations that arrive after targeting. At best, it replaces contemporaneous predictors with staler ones. Figure 6c is consistent with exactly that prediction. The PMT trained to predict future consumption performs worse than the standard PMT. Taken together, these results support the model’s characterization of poverty dynamics: the static PMT captures chronic poverty reasonably well at baseline, but it cannot prevent targeting error from rising over time as chronic poverty evolves. The remaining question is whether a dynamic PMT can limit that increase in error, and at what cost.

5.2. Updating PMTs over time

Updating PMTs can reduce the increase in targeting error over time, but only to the extent that updated information captures changes in chronic poverty. Consistent with the model, modest updating helps somewhat, while fully redoing the targeting exercise is the only way to substantially limit the deterioration caused by poverty dynamics.

The extension of the model shows that dynamic PMTs differ in which source of chronic-poverty error they can reduce, and therefore in whether their gains are likely to justify their cost. Appendix D.2 formalizes three updating schemes. Proposition 9 shows that re-estimating the PMT formula alone cannot help much when the characteristics of the population remain stale, because any improvement in fit in the training sample is applied to outdated household information. Proposition 8 shows that updating predictors while holding the formula fixed can help if observed characteristics track changes in chronic poverty and the mapping from characteristics to chronic poverty remains reasonably stable. Proposition 10 shows that updating both the formula and the predictors is the only way to remove the mechanical deterioration caused by predictable changes in chronic poverty, though it remains unable to track transitory poverty. The empirical results below therefore ask which of these margins matters enough in practice to justify the cost of redoing part, or all, of the PMT exercise.

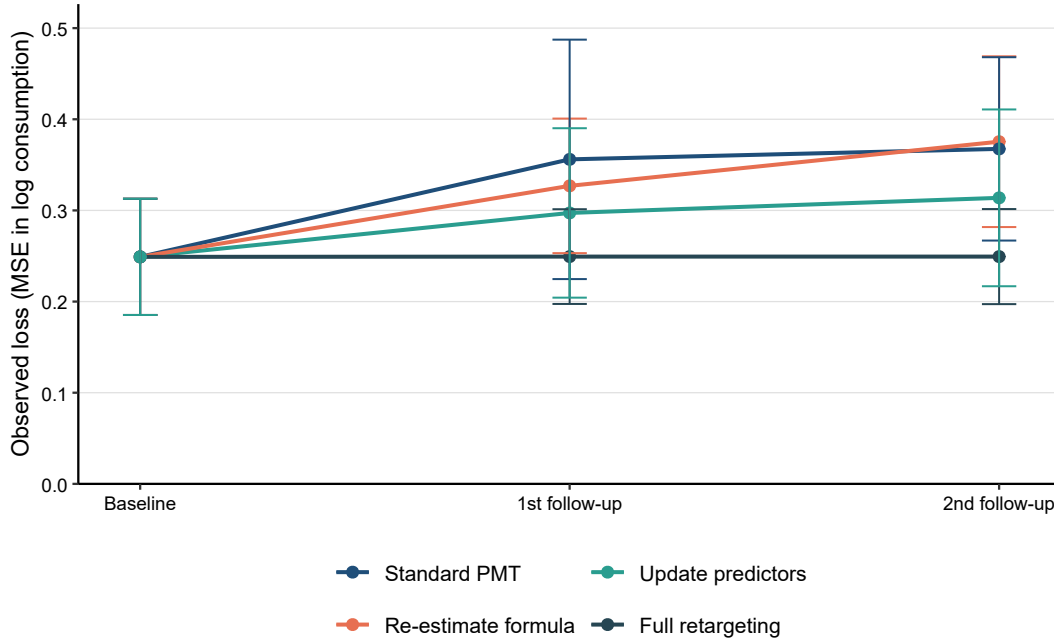
Re-estimating the PMT formula alone yields little improvement, confirming the model’s prediction that stale population characteristics are the binding constraint. In Figure 7, the line for “Re-estimate formula” lies very close to the standard PMT. This is exactly the pattern predicted by Proposition 9: better fit in the training sample does not translate into better targeting in the population when household rankings are still based on old characteristics. This scheme is relatively cheap because it requires new consumption and characteristics only in the training sample, not in the broader population. But precisely because it does not update the information used to rank households, it delivers little value for money. The natural next question is therefore whether it is more useful to spend

resources on updating the population predictors themselves.

Updating population predictors while keeping the PMT formula fixed yields a meaningful, but still partial, improvement. In Figure 7, the line for “Update predictors” rises much more slowly than the standard PMT, especially by period 2. This is consistent with Proposition 8: updated predictors allow the PMT to track the part of chronic-poverty change that becomes visible in household characteristics, although keeping the formula fixed still leaves error when the mapping from characteristics to chronic poverty changes over time. Compared with full retargeting, this scheme saves the cost of collecting new consumption data for a training sample and recalibrating the PMT. But it still requires remeasuring predictors in the full population, which is itself expensive. The remaining question is whether the additional cost of fully redoing the PMT is justified by a large further gain in targeting precision.

Fully redoing the PMT yields the largest improvement, but it is also the most expensive option. In Figure 7, “Full retargeting” keeps targeting error essentially flat over time and clearly below all other PMT variants. This is exactly what Proposition 10 predicts: once both the PMT formula and the population predictors are updated, the PMT no longer mechanically deteriorates because of changes in the predictable part of chronic poverty. The remaining targeting error is chronic-poverty heterogeneity not captured by observables and, if the target were current poverty, transitory poverty, plus measurement error for the observed targeting error. The policy implication is therefore not that full retargeting should always be adopted, but that it is the only PMT design that can substantially offset the increase in targeting error over time. Whether it is worth the cost depends on the size of the program, the cost of targeting errors, and the administrative cost of recollecting both training data and population predictors.

Figure 7 — Updating PMTs over time



Notes: The figure compares the observed performance over time of the standard PMT to three updating strategies. The standard PMT keeps both the baseline formula and the baseline predictors fixed over time. “Update predictors” updates household predictors while keeping the original PMT formula fixed. “Re-estimate formula” updates the PMT formula using new training data while keeping baseline predictors fixed in the population. “Full retargeting” repeats the full targeting exercise. The outcome is observed mean squared error, averaged across the ten low-frequency country panels and 100 repeated train-test splits. Error bars report 95% confidence intervals across country panels, after averaging over repeated splits within each country.

A final possibility is to update the PMT using cheap, high-frequency information such as climate or satellite-based predictors. In principle, this is attractive because it offers the possibility of more frequent updates at low cost. In practice, however, Appendix Figure D.1 suggests that this margin is hard to exploit. We test such updates in Ethiopia using meteorological variables and aggregate community characteristics as proxies for information that could be updated cheaply over time. The gains are limited. This is consistent with the broader logic of the model: updating only a subset of predictors is unlikely to help much when the PMT function and the remaining household-level information remain anchored in the past. Cheap dynamic inputs may still be useful in specific contexts, but our results suggest that they are not an easy substitute for remeasuring the predictors that matter most for chronic-poverty targeting.

5.3. Design lessons for policy

The results in this section suggest a simple policy rule. If the goal is to target chronic poverty, the standard PMT performs reasonably well, and the static refinements we study do not yield robust improvements. PMTs are therefore best understood as a relatively low-cost tool for chronic-poverty targeting. The main design dimension is then whether

the expected gains from updating — that is, from better tracking changes in chronic poverty over time — justify the administrative cost.

The answer is different when the policy objective is to capture current need. Current poverty includes transitory poverty, which PMTs do not target by design. More generally, even a static method that did target transitory poverty at calibration would not track current poverty well over time, because transitory shocks are replaced by new shocks after targeting. Programs such as temporary relief or shock response therefore require targeting methods that are designed to capture current poverty and that are updated frequently enough to keep pace with changing need.

6. Conclusion

Many anti-poverty programs must choose beneficiary lists using noisy and infrequent information about household welfare, and then keep those lists fixed for some time. The standard evaluation approach asks whether a targeting rule matches observed poverty at calibration. This paper argues that this is the wrong object for evaluating targeting over time. A beneficiary list is used during program implementation, not only at calibration, and observed poverty combines persistent deprivation, temporary shocks, and measurement error. These components have different implications for targeting: measurement error should not be targeted, temporary shocks are hard for any static rule to track, and changes in persistent living standards determine how quickly static targeting rules become stale.

The framework connects literatures that usually remain separate. The poverty-dynamics literature distinguishes chronic from transitory poverty (Jalan and Ravallion, 2000; Duclos et al., 2010; Dang and Dabalen, 2019), while the poverty-measurement literature shows that observed welfare is measured with error (Ravallion, 2008; Glewwe, 2012). Our contribution is to bring these distinctions into the targeting problem. A program that aims to respond to temporary shocks and a program that aims to reach persistently poor households are not asking the same targeting question. They should therefore not be evaluated against the same static observed-poverty benchmark.

Empirically, the distinction matters. Across ten low-frequency country panels, the observed-poverty benchmark has zero error at calibration by construction, but bottom-30 classification error rises to 25.1 percent by the first follow-up and 26.8 percent by the second. In the monthly India panel, targeting error jumps to 11.7 percent after only one month and reaches 23.4 percent by month 12. Mapping these dynamics to the framework shows that persistent changes in living standards account for only 19 percent of the initial observed movement in poverty, while transitory shocks and measurement error account for 81 percent. This is why static targeting can look worse than it is at baseline, but also why no static rule should be expected to track all movements in current poverty.

This perspective changes the interpretation of Proxy Means Tests. PMTs are often judged against observed poverty and therefore appear to make large errors. But their original informational logic was different: when income or consumption is costly to observe directly, observable household characteristics provide proxies for longer-run welfare (Ravallion and Chao, 1989; Glewwe et al., 1989; Grosh and Baker, 1995). This logic is also consistent with standard consumption theory, in which consumption is tied to permanent resources

rather than purely temporary shocks (Deaton, 1992; Jappelli and Pistaferri, 2017). We show formally that, under the maintained assumptions, PMTs predict the component of consumption encoded in household characteristics, not transitory shocks or measurement error. In the data, PMT classification error rises only gradually, from 21.9 percent at calibration to 24.9 percent at the first follow-up and 26.6 percent at the second. Moreover, 70 percent of observed baseline PMT loss reflects transitory poverty and measurement error rather than mistargeting of persistent deprivation.

The same logic explains why attempts to improve static targeting rules deliver limited gains. Recent advances in prediction and data availability have encouraged efforts to reduce observed targeting error using machine learning, alternative training targets, and richer data (Kleinberg et al., 2015; Mullainathan and Spiess, 2017; McBride and Nichols, 2016; Kshirsagar et al., 2017; Areias and Wai-Poi, 2022). These tools may improve prediction in some settings, but they cannot remove the part of observed error that comes from measurement error, temporary shocks, or future changes in living standards. In our data, machine learning PMTs remain close to standard PMTs, PMTs trained on averaged consumption perform almost identically to the standard PMT, and PMTs trained to forecast future poverty perform worse. The binding problem is that a static rule cannot target shocks or innovations that have not yet occurred.

The results also clarify the debate over how often targeting rules should be updated. Recent work documents that targeting performance deteriorates as data age and argues that more recent data can improve targeting accuracy (Aiken et al., 2023; Hillebrecht et al., 2023; Beuermann et al., 2025; Chia and George, 2026). Our results support the premise that stale data matter, but they also imply a more nuanced policy conclusion. Frequent retargeting is not costless. Beyond the direct expense of collecting new data, it can increase administrative burden, create uncertainty for households, and repeatedly move households into and out of programs. The relevant question is therefore not whether updating mechanically improves accuracy, but whether the gain is worth those costs given the program’s objective. If the goal is to respond to temporary shocks or current need, frequent updating or a separate shock-response system is necessary: households should be included when shocks hit and excluded once those shocks pass. If the goal is to reach persistent deprivation, static PMTs can perform reasonably well and deteriorate slowly, so updating should occur when the expected gain in precision justifies the cost.

The broader lesson is that targeting evaluations should be built around the poverty object the program is meant to reach. Observed poverty at one point in time is a useful measurement input, but it is not the same as current need, persistent deprivation, or vulnerability. Once this distinction is made, PMTs look less like failed predictors of observed poverty and more like stable predictors of one important component of welfare. At the same time, their limits become clearer: static tools can help reach households facing persistent disadvantage, but programs that aim to insure households against changing shocks require dynamic information. Aligning targeting evaluation with program objectives is therefore the first step toward designing targeting systems that are both accurate and administratively sensible.

References

- Aiken, Emily, Tim Ohlenburg, and Joshua Blumenstock.** 2023. “Moving Targets: The Role of Model and Data Recency in Proxy Means Test Accuracy.” August, Working paper. [Cited on pages 5 and 29.]
- Alatas, Vivi, Abhijit Banerjee, Rema Hanna, Benjamin A Olken, and Julia Tobias.** 2012. “Targeting the poor: evidence from a field experiment in Indonesia.” *American Economic Review* 102 (4): 1206–40. [Cited on page 4.]
- Areias, Ana, and Matthew Wai-Poi.** 2022. “Big Data, Machine Learning and Prediction in the Targeting of Social Protection Programs.” In *Revisiting Targeting in Social Assistance: A New Look at Old Dilemmas*, edited by Grosh, Margaret, Phillippe Leite, Matthew Wai-Poi, and Emil Tesliuc, Washington, DC: World Bank. [Cited on pages 4 and 29.]
- Armentano, Vincent, Paul Niehaus, and Tom Vogl.** 2026. “How Poverty Fell.” Technical report, University of California, San Diego. [Cited on pages 1 and 6.]
- Beuermann, Diether W., Bridget Hoffmann, Marco Stampini, David L. Vargas, and Diego Vera-Cossio.** 2025. “Shooting a Moving Target: Evaluating Targeting Tools for Social Programs When Income Fluctuates.” *Journal of Development Economics* 172 103395. [10.1016/j.jdeveco.2024.103395](https://doi.org/10.1016/j.jdeveco.2024.103395). [Cited on pages 5 and 29.]
- Blundell, Richard, Luigi Pistaferri, and Ian Preston.** 2008. “Consumption inequality and partial insurance.” *American Economic Review* 98 (5): 1887–1921. [Cited on pages 2 and 6.]
- Brown, Caitlin, Martin Ravallion, and Dominique Van de Walle.** 2018. “A poor means test? Econometric targeting in Africa.” *Journal of Development Economics* 134 109–124. [Cited on page 4.]
- Chaudhuri, Shubham, and Martin Ravallion.** 1994. “How Well Do Static Indicators Identify the Chronically Poor?” *Journal of Public Economics* 53 (3): 367–394. [10.1016/0047-2727\(94\)90031-0](https://doi.org/10.1016/0047-2727(94)90031-0). [Cited on page 2.]
- Chesher, Andrew, and Christian Schluter.** 2002. “Welfare Measurement and Measurement Error.” *The Review of Economic Studies* 69 (2): 357–378. [10.1111/1467-937X.00210](https://doi.org/10.1111/1467-937X.00210). [Cited on page 2.]
- Chia, Rafi, and Siddharth George.** 2026. “Chasing Yesterday’s Poor: The Limits of Targeting in a Volatile World.” April, Working paper. [Cited on pages 5 and 29.]
- Coady, David, Margaret Grosh, and John Hoddinott.** 2004. *Targeting of transfers in developing countries: Review of lessons and experience*. The World Bank. [Cited on page 4.]
- Collins, Daryl, Jonathan Morduch, Stuart Rutherford, and Orlanda Ruthven.** 2009. *Portfolios of the poor: how the world’s poor live on 2aday*. Princeton University Press. [Cited on pages 1 and 6.]

- Dang, Hai-Anh H, and Andrew L Dabalen.** 2019. “Is poverty in Africa mostly chronic or transient? Evidence from synthetic panel data.” *The Journal of Development Studies* 55 (7): 1527–1547. [Cited on pages 3 and 28.]
- Deaton, Angus.** 1992. *Understanding consumption*. Oxford University Press. [Cited on pages 2, 4, 9, and 29.]
- Deaton, Angus.** 1997. *The analysis of household surveys: a microeconomic approach to development policy*. World Bank Publications. [Cited on page 14.]
- Deaton, Angus, and Salman Zaidi.** 2002. *Guidelines for constructing consumption aggregates for welfare analysis*. Volume 135. World Bank Publications. [Cited on page 14.]
- Del Ninno, Carlo, and Bradford Mills.** 2015. *Safety Nets in Africa: Effective Mechanisms to Reach the Poor and Most Vulnerable*. The World Bank. [Cited on page 3.]
- Devereux, Stephen, Edoardo Masset, Rachel Sabates-Wheeler, Michael Samson, Althea-Maria Rivas, and Dolf Te Lintelo.** 2017. “The targeting effectiveness of social transfers.” *Journal of Development Effectiveness* 9 (2): 162–211. [Cited on page 4.]
- Duclos, Jean-Yves, Abdelkrim Araar, and John Giles.** 2010. “Chronic and transient poverty: Measurement and estimation, with evidence from China.” *Journal of development Economics* 91 (2): 266–277. [Cited on pages 2, 3, and 28.]
- Glewwe, Paul.** 2012. “How Much of Observed Economic Mobility is Measurement Error? IV Methods to Reduce Measurement Error Bias, with an Application to Vietnam.” *The World Bank Economic Review* 26 (2): 236–264. [10.1093/wber/lhr040](https://doi.org/10.1093/wber/lhr040). [Cited on pages 2, 3, and 28.]
- Glewwe, Paul, Oussama Kanaan et al.** 1989. *Targeting assistance to the poor using household survey data*. Volume 225. World Bank. [Cited on pages 4 and 28.]
- Grosh, Margaret, Carlo Del Ninno, Emil Tesliuc, and Azedine Ouerghi.** 2008. *For protection and promotion: The design and implementation of effective safety nets*. World Bank Publications. [Cited on page 3.]
- Grosh, Margaret E., and Judy L. Baker.** 1995. “Proxy Means Tests for Targeting Social Programs: Simulations and Speculation.” Working Paper 118, Living Standards Measurement Study, World Bank. [Cited on pages 4 and 28.]
- Grosh, Margaret, and Paul Glewwe.** 2000. *Designing household survey questionnaires for developing countries*. Washington, DC: World Bank. [Cited on page 14.]
- Grosh, Margaret, Phillippe Leite, Matthew Wai-Poi, and Emil Tesliuc.** 2022. *Revisiting targeting in social assistance: A new look at old dilemmas*. World Bank Publications. [Cited on page 3.]
- Hall, Robert E.** 1978. “Stochastic implications of the life cycle-permanent income hypothesis: theory and evidence.” *Journal of political economy* 86 (6): 971–987. [Cited on page 2.]

- Hillebrecht, Michael, Stefan Klonner, and Noraogo A Pacere.** 2023. “The dynamics of poverty targeting.” *Journal of Development Economics* 161 103033. [Cited on pages 5 and 29.]
- Jalan, Jyotsna, and Martin Ravallion.** 1998. “Transient Poverty in Postreform Rural China.” *Journal of Comparative Economics* 26 (2): 338–357. [10.1006/jcec.1998.1526](https://doi.org/10.1006/jcec.1998.1526). [Cited on page 3.]
- Jalan, Jyotsna, and Martin Ravallion.** 2000. “Is Transient Poverty Different? Evidence for Rural China.” *The Journal of Development Studies* 36 (6): 82–99. [10.1080/00220380008422655](https://doi.org/10.1080/00220380008422655). [Cited on pages 2, 3, and 28.]
- Jappelli, Tullio, and Luigi Pistaferri.** 2017. *The economics of consumption: theory and evidence*. Oxford University Press. [Cited on pages 4, 9, and 29.]
- Kidd, Stephen, and Emily Wylde.** 2011. *Targeting the Poorest: An Assessment of the Proxy Means Test Methodology*. Canberra: Australian Agency for International Development (AusAID). [Cited on page 4.]
- Kleinberg, Jon, Jens Ludwig, Sendhil Mullainathan, and Ziad Obermeyer.** 2015. “Prediction policy problems.” *American Economic Review* 105 (5): 491–95. [Cited on page 29.]
- Kshirsagar, Varun, Jerzy Wiecezorek, Sharada Ramanathan, and Rachel Wells.** 2017. “Household poverty classification in data-scarce environments: a machine learning approach.” *arXiv preprint arXiv:1711.06813*. [Cited on pages 4 and 29.]
- Lee, Nayoung, Geert Ridder, and John Strauss.** 2017. “Estimation of Poverty Transition Matrices with Noisy Data.” *Journal of Applied Econometrics* 32 (1): 37–55. [10.1002/jae.2506](https://doi.org/10.1002/jae.2506). [Cited on page 3.]
- McBride, Linden, and Austin Nichols.** 2016. “Retooling poverty targeting using out-of-sample validation and machine learning.” *The World Bank Economic Review* 32 (3): 531–550. [Cited on pages 4 and 29.]
- Merfeld, Joshua D., and Jonathan Morduch.** 2026. “Poverty at Higher Frequency.” February, Working paper. [Cited on pages 1 and 6.]
- Mullainathan, Sendhil, and Jann Spiess.** 2017. “Machine learning: an applied econometric approach.” *Journal of Economic Perspectives* 31 (2): 87–106. [Cited on page 29.]
- Paxson, Christina H.** 1992. “Using Weather Variability to Estimate the Response of Savings to Transitory Income in Thailand.” *The American Economic Review* 82 (1): 15–33. [Cited on page 4.]
- Premad, Patrick, and Pascale Schnitzer.** 2018. *Efficiency, legitimacy and impacts of targeting methods: Evidence from an experiment in Niger*. The World Bank. [Cited on page 4.]
- Ravallion, Martin.** 2008. “Miss-Targeted, or Miss-Measured?” *Economics Letters* 100 9–12. [Cited on page 28.]

- Ravallion, Martin.** 2016. *The economics of poverty: History, measurement, and policy*. Oxford University Press. [Cited on page 14.]
- Ravallion, Martin, and Calvin Chao.** 1989. “Targeted policies for poverty alleviation under imperfect information: algorithms and applications.” *Journal of Policy modeling* 11 (2): 213–224. [Cited on pages 4 and 28.]
- Santaeulalia-Llopis, Raül, and Yu Zheng.** 2018. “The price of growth: consumption insurance in China 1989–2009.” *American Economic Journal: Macroeconomics* 10 (4): 1–35. [Cited on page 6.]
- Sumarto, Sudarno, Asep Suryahadi, and Lant Pritchett.** 2003. “Safety Nets or Safety Ropes? Dynamic Benefit Incidence of Two Crisis Programs in Indonesia.” *World Development* 31 (7): 1257–1277. [Cited on page 5.]
- van de Walle, Dominique.** 2002. “The Static and Dynamic Incidence of Vietnam’s Public Safety Net.” Technical Report 2791, World Bank, Washington, DC. [Cited on page 5.]

Appendix

A. Framework derivations

A.1. Consumption decomposition

Define permanent consumption as expected next-period consumption,

$$c_{it}^P \equiv \mathbb{E}_t[c_{i,t+1}],$$

and transitory consumption as the deviation from that path,

$$c_{it}^T \equiv c_{it} - c_{it}^P.$$

Lemma 1 (Permanent consumption path). Under the income process above and the reduced-form consumption rule

$$c_{it} = p_{it} + \alpha u_{it},$$

permanent consumption coincides with permanent income:

$$c_{it}^P = p_{it}.$$

Proof. Using the reduced-form consumption rule,

$$c_{i,t+1} = p_{i,t+1} + \alpha u_{i,t+1}.$$

Taking expectations conditional on information at time t gives

$$\begin{aligned} \mathbb{E}_t[c_{i,t+1}] &= \mathbb{E}_t[p_{i,t+1}] + \alpha \mathbb{E}_t[u_{i,t+1}] \\ &= \mathbb{E}_t[p_{it} + \eta_{i,t+1}] + \alpha \cdot 0 \\ &= p_{it}, \end{aligned}$$

where the second line uses the martingale property of permanent income and the zero conditional mean of transitory shocks. Hence

$$c_{it}^P = \mathbb{E}_t[c_{i,t+1}] = p_{it}.$$

□

Corollary 2 (Transitory consumption). Under the same assumptions, transitory consumption is

$$c_{it}^T = \alpha u_{it}.$$

Proof. By definition,

$$c_{it}^T = c_{it} - c_{it}^P.$$

Substituting $c_{it} = p_{it} + \alpha u_{it}$ and $c_{it}^P = p_{it}$ yields

$$c_{it}^T = (p_{it} + \alpha u_{it}) - p_{it} = \alpha u_{it}.$$

□

The two polar cases follow immediately. In the hand-to-mouth case ($\alpha = 1$), transitory consumption equals the full transitory shock:

$$c_{it}^T = u_{it}.$$

In the strict permanent-income case ($\alpha = 0$), transitory consumption is zero:

$$c_{it}^T = 0.$$

Combining the lemma and corollary yields the decomposition used in the main text:

$$c_{it} = c_{it}^P + c_{it}^T = p_{it} + \alpha u_{it}.$$

A.2. Status-quo bias for static targeting rules

This subsection derives the bias of the standard calibration-period evaluation for a generic static targeting rule. Let $\hat{c}_{i0}^f$ denote the score assigned at baseline by rule f . We maintain the assumptions from the main text: permanent innovations, transitory shocks, and measurement error have mean zero; future innovations and future transitory shocks are orthogonal to baseline quantities; and classical measurement error is orthogonal to true consumption and its components. We focus first on rules that do not mechanically target baseline measurement error:

$$\mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)\omega_{i0}] = 0.$$

The rule may, however, load on baseline transitory consumption. Define θ_f by

$$\mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)u_{i0}] = -\theta_f \alpha \sigma_u^2.$$

Thus $\theta_f = 0$ for a rule that targets only persistent deprivation, while larger values indicate that the rule also captures baseline transitory consumption. For horizons $t > 0$, evaluation-period permanent innovations, transitory shocks, and measurement error are newly realized after the baseline score is fixed.

Proposition 1 (Bias of the standard evaluation criterion). For any static targeting rule f satisfying the no-measurement-error condition above, the standard calibration-period loss is

$$\mathcal{L}_0^{obs}(f) = \mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)^2] + (1 - 2\theta_f)\alpha^2\sigma_u^2 + \sigma_\omega^2.$$

For any horizon $t > 0$, policy-relevant loss for current-poverty targeting is

$$\mathcal{L}_t(f; c) = \mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)^2] + t\sigma_\eta^2 + \alpha^2\sigma_u^2,$$

and policy-relevant loss for chronic-poverty targeting is

$$\mathcal{L}_t(f; c^P) = \mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)^2] + t\sigma_\eta^2.$$

Therefore,

$$\mathcal{L}_0^{obs}(f) - \mathcal{L}_t(f; c) = \sigma_\omega^2 - t\sigma_\eta^2 - 2\theta_f\alpha^2\sigma_u^2,$$

and

$$\mathcal{L}_0^{obs}(f) - \mathcal{L}_t(f; c^P) = \sigma_\omega^2 + (1 - 2\theta_f)\alpha^2\sigma_u^2 - t\sigma_\eta^2.$$

Proof. For the standard calibration-period loss,

$$\tilde{c}_{i0} - \hat{c}_{i0}^f = p_{i0} - \hat{c}_{i0}^f + \alpha u_{i0} + \omega_{i0}.$$

Squaring and taking expectations gives

$$\begin{aligned} \mathcal{L}_0^{obs}(f) &= \mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)^2] + \alpha^2\sigma_u^2 + \sigma_\omega^2 + 2\alpha\mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)u_{i0}] + 2\mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)\omega_{i0}] + 2\alpha\mathbb{E}[u_{i0}\omega_{i0}] \\ &= \mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)^2] + (1 - 2\theta_f)\alpha^2\sigma_u^2 + \sigma_\omega^2. \end{aligned}$$

The second equality uses the no-measurement-error condition, classical measurement-error orthogonality, and the definition of θ_f .

For $t > 0$, current consumption is

$$c_{it} = p_{i0} + \sum_{s=1}^t \eta_{is} + \alpha u_{it}.$$

Thus

$$c_{it} - \hat{c}_{i0}^f = p_{i0} - \hat{c}_{i0}^f + \sum_{s=1}^t \eta_{is} + \alpha u_{it}.$$

Future permanent innovations and future transitory shocks have mean zero and do not move systematically with the baseline score. Cross terms therefore vanish in expectation, and

$$\mathcal{L}_t(f; c) = \mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)^2] + \text{Var}\left(\sum_{s=1}^t \eta_{is}\right) + \alpha^2\sigma_u^2 = \mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)^2] + t\sigma_\eta^2 + \alpha^2\sigma_u^2.$$

For chronic poverty, $c_{it}^P = p_{i0} + \sum_{s=1}^t \eta_{is}$, so the same argument gives

$$\mathcal{L}_t(f; c^P) = \mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)^2] + t\sigma_\eta^2.$$

Subtracting these two policy-relevant losses from $\mathcal{L}_0^{obs}(f)$ gives the stated bias expressions. $\square$

Corollary 3 (Observed and policy-relevant loss after calibration). For horizons $t > 0$, observed loss is

$$\mathcal{L}_t^{obs}(f) = \mathbb{E}[(p_{i0} - \hat{c}_{i0}^f)^2] + t\sigma_\eta^2 + \alpha^2\sigma_u^2 + \sigma_\omega^2.$$

It therefore exceeds current-poverty loss by measurement error,

$$\mathcal{L}_t^{obs}(f) - \mathcal{L}_t(f; c) = \sigma_\omega^2,$$

and exceeds chronic-poverty loss by transitory poverty plus measurement error,

$$\mathcal{L}_t^{obs}(f) - \mathcal{L}_t(f; c^P) = \alpha^2 \sigma_u^2 + \sigma_\omega^2.$$

The same expressions also hold at calibration for rules that do not mechanically target baseline transitory consumption or baseline measurement error.

A.3. Proxy-Means Tests target chronic consumption

Lemma 4 (PMTs target permanent consumption). Suppose permanent income is a function of observed household characteristics Γ_i and unobserved characteristics X_i , and observed consumption equals permanent consumption plus transitory shocks and classical measurement error. Assume that transitory shocks and measurement error are orthogonal to observed household characteristics. Then the population PMT coefficient obtained by regressing observed consumption on observed household characteristics identifies the linear projection of permanent income on those characteristics. In particular, transitory shocks and measurement error do not affect the PMT slope coefficient. As a result, PMTs target chronic (permanent) poverty rather than transitory consumption fluctuations.

Proof. Suppose permanent income is a function of observed household characteristics Γ_i and unobserved characteristics X_i , $p_i = g(\Gamma_i, X_i)$. Consistent with the income process above, transitory shocks are assumed to be independent of household characteristics, and classical measurement error is orthogonal to household characteristics. Consider the linear projection of permanent income onto observed household characteristics, assumed to be a scalar random variable, in the cross section:

$$p_i = \beta_0 + \beta \Gamma_i + \varepsilon_i,$$

where $\text{Cov}(\Gamma_i, \varepsilon_i) = 0$. Consider the population regression used to construct a PMT,

$$\tilde{c}_i = \beta_0^{PMT} + \beta^{PMT} \Gamma_i + e_i.$$

The population OLS slope in the PMT regression is

$$\beta^{PMT} = \frac{\text{Cov}(\tilde{c}_i, \Gamma_i)}{\text{Var}(\Gamma_i)}.$$

Using $\tilde{c}_i = p_i + \alpha u_i + \omega_i$, we obtain

$$\begin{aligned} \text{Cov}(\tilde{c}_i, \Gamma_i) &= \text{Cov}(p_i + \alpha u_i + \omega_i, \Gamma_i) \\ &= \text{Cov}(p_i, \Gamma_i) + \alpha \text{Cov}(u_i, \Gamma_i) + \text{Cov}(\omega_i, \Gamma_i). \end{aligned}$$

By assumption, $\text{Cov}(u_i, \Gamma_i) = 0$ and $\text{Cov}(\omega_i, \Gamma_i) = 0$, hence

$$\text{Cov}(\tilde{c}_i, \Gamma_i) = \text{Cov}(p_i, \Gamma_i).$$

Using the linear projection $p_i = \beta_0 + \beta\Gamma_i + \varepsilon_i$ and $\text{Cov}(\Gamma_i, \varepsilon_i) = 0$,

$$\begin{aligned}\text{Cov}(p_i, \Gamma_i) &= \text{Cov}(\beta_0 + \beta\Gamma_i + \varepsilon_i, \Gamma_i) \\ &= \beta \text{Var}(\Gamma_i).\end{aligned}$$

Therefore,

$$\beta^{PMT} = \frac{\text{Cov}(\tilde{c}_i, \Gamma_i)}{\text{Var}(\Gamma_i)} = \frac{\beta \text{Var}(\Gamma_i)}{\text{Var}(\Gamma_i)} = \beta.$$

□

Corollary 5 (Interpretation). The regression residual aggregates the components of consumption that are not predictable from observed characteristics:

$$e_i = \varepsilon_i + \alpha u_i + \omega_i.$$

Thus, PMTs miss (i) permanent heterogeneity not captured by observed characteristics (ε_i), (ii) transitory consumption fluctuations (αu_i), and (iii) measurement error (ω_i).

Proof. Substituting the economic structure into the PMT regression yields

$$\begin{aligned}\tilde{c}_i &= \beta_0^{PMT} + \beta^{PMT}\Gamma_i + e_i, \\ p_i + \alpha u_i + \omega_i &= \beta_0^{PMT} + \beta^{PMT}\Gamma_i + e_i.\end{aligned}$$

Using the projection of permanent income and $\beta^{PMT} = \beta$,

$$\beta_0 + \beta\Gamma_i + \varepsilon_i + \alpha u_i + \omega_i = \beta_0^{PMT} + \beta\Gamma_i + e_i.$$

The intercept in the PMT regression equals the intercept in the permanent-income projection, $\beta_0^{PMT} = \beta_0$, because $\mathbb{E}[\varepsilon_i] = \mathbb{E}[u_i] = \mathbb{E}[\omega_i] = 0$. Deriving the PMT regression residual therefore yields the decomposition:

$$e_i = \varepsilon_i + \alpha u_i + \omega_i.$$

□

A.4. Loss paths of proxy-means tests

We now specialize the static-rule results to PMTs. Appendix A.3 shows that, under the maintained assumptions, the PMT score predicts the component of chronic consumption encoded in household characteristics. Let

$$\varepsilon_{i0} \equiv p_{i0} - \hat{c}_{i0}^{PMT}$$

denote the remaining chronic-prediction error, with $\sigma_\varepsilon^2 \equiv \text{Var}(\varepsilon_{i0})$. The key assumption is not that PMTs perfectly predict chronic poverty; they do not. The assumption is that the PMT score does not mechanically target the realized transitory shock or measurement error in the consumption survey:

$$\mathbb{E}[\varepsilon_{i0}u_{i0}] = 0, \quad \mathbb{E}[\varepsilon_{i0}\omega_{i0}] = 0.$$

This follows from the fact that PMTs use household characteristics rather than the realized consumption residual itself.

Corollary 6 (Loss paths of proxy-means tests). Under the model in Section 2, the PMT loss paths for all horizons $t \geq 0$ are

$$\mathcal{L}_t(\hat{c}_{i0}^{PMT}; c) = t\sigma_\eta^2 + \alpha^2\sigma_u^2 + \sigma_\varepsilon^2,$$

$$\mathcal{L}_t^{obs}(\hat{c}_{i0}^{PMT}) = t\sigma_\eta^2 + \alpha^2\sigma_u^2 + \sigma_\omega^2 + \sigma_\varepsilon^2,$$

and, for chronic-poverty targeting,

$$\mathcal{L}_t(\hat{c}_{i0}^{PMT}; c^P) = t\sigma_\eta^2 + \sigma_\varepsilon^2.$$

Proof. For a PMT, $p_{i0} - \hat{c}_{i0}^{PMT} = \varepsilon_{i0}$, so $\mathbb{E}[(p_{i0} - \hat{c}_{i0}^{PMT})^2] = \sigma_\varepsilon^2$. Under the PMT orthogonality condition, $\theta_{PMT} = 0$. Substituting into Proposition 1 gives the current- and chronic-poverty loss paths for $t > 0$, and Corollary 3 gives the observed loss path. Because the PMT score does not mechanically include realized transitory shocks or measurement error, the same expressions apply at calibration as well. At calibration, $c_{i0} - \hat{c}_{i0}^{PMT} = \varepsilon_{i0} + \alpha u_{i0}$, $\tilde{c}_{i0} - \hat{c}_{i0}^{PMT} = \varepsilon_{i0} + \alpha u_{i0} + \omega_{i0}$, and $c_{i0}^P - \hat{c}_{i0}^{PMT} = \varepsilon_{i0}$, which gives the same expressions with $t = 0$. $\square$

A.5. Loss paths of the baseline observed-poverty benchmark

We now specialize to the baseline observed-poverty benchmark, which uses $\tilde{c}_{i0}$ itself as the targeting score. This benchmark is not covered by the generic result at calibration because its score contains baseline transitory consumption and baseline measurement error.

Lemma 7 (Loss paths of the baseline observed-poverty benchmark). Under the model in Section 2, the true and observed losses of the baseline observed-poverty benchmark are:

$$\mathcal{L}_t(\tilde{c}_{i0}; c) = \begin{cases} \sigma_\omega^2, & \text{for } t = 0, \\ t\sigma_\eta^2 + 2\alpha^2\sigma_u^2 + \sigma_\omega^2, & \text{for } t > 0, \end{cases}$$

$$\mathcal{L}_t(\tilde{c}_{i0}; c^P) = \begin{cases} \alpha^2\sigma_u^2 + \sigma_\omega^2, & \text{for } t = 0, \\ t\sigma_\eta^2 + \alpha^2\sigma_u^2 + \sigma_\omega^2, & \text{for } t > 0, \end{cases}$$

and

$$\mathcal{L}_t^{obs}(\tilde{c}_{i0}) = \begin{cases} 0, & \text{for } t = 0, \\ t\sigma_\eta^2 + 2\alpha^2\sigma_u^2 + 2\sigma_\omega^2, & \text{for } t > 0. \end{cases}$$

Proof. Using $\tilde{c}_{i0} = p_{i0} + \alpha u_{i0} + \omega_{i0}$ and $c_{it} = p_{i0} + \sum_{s=1}^t \eta_{is} + \alpha u_{it}$, current-poverty loss is

$$c_{it} - \tilde{c}_{i0} = \sum_{s=1}^t \eta_{is} + \alpha(u_{it} - u_{i0}) - \omega_{i0}.$$

At $t = 0$, this reduces to $-\omega_{i0}$, so the loss is σ_ω^2 . For $t > 0$, the maintained orthogonality assumptions imply that the cross terms are zero, so

$$\mathcal{L}_t(\tilde{c}_{i0}; c) = \text{Var}\left(\sum_{s=1}^t \eta_{is}\right) + \alpha^2 \text{Var}(u_{it} - u_{i0}) + \text{Var}(\omega_{i0}) = t\sigma_\eta^2 + 2\alpha^2\sigma_u^2 + \sigma_\omega^2.$$

For chronic-poverty loss, use $c_{it}^P = p_{it} = p_{i0} + \sum_{s=1}^t \eta_{is}$. Then

$$c_{it}^P - \tilde{c}_{i0} = \sum_{s=1}^t \eta_{is} - \alpha u_{i0} - \omega_{i0}.$$

At $t = 0$, the loss is $\alpha^2\sigma_u^2 + \sigma_\omega^2$. For $t > 0$,

$$\mathcal{L}_t(\tilde{c}_{i0}; c^P) = \text{Var}\left(\sum_{s=1}^t \eta_{is}\right) + \alpha^2 \text{Var}(u_{i0}) + \text{Var}(\omega_{i0}) = t\sigma_\eta^2 + \alpha^2\sigma_u^2 + \sigma_\omega^2.$$

For observed loss, use $\tilde{c}_{it} = p_{i0} + \sum_{s=1}^t \eta_{is} + \alpha u_{it} + \omega_{it}$. Then

$$\tilde{c}_{it} - \tilde{c}_{i0} = \sum_{s=1}^t \eta_{is} + \alpha(u_{it} - u_{i0}) + (\omega_{it} - \omega_{i0}).$$

At $t = 0$, observed loss is zero mechanically. For $t > 0$,

$$\mathcal{L}_t^{obs}(\tilde{c}_{i0}) = \text{Var}\left(\sum_{s=1}^t \eta_{is}\right) + \alpha^2 \text{Var}(u_{it} - u_{i0}) + \text{Var}(\omega_{it} - \omega_{i0}) = t\sigma_\eta^2 + 2\alpha^2\sigma_u^2 + 2\sigma_\omega^2.$$

□

A.6. Low-frequency identification of the transitory poverty and measurement error wedge

Let

$$W \equiv \alpha^2\sigma_u^2 + \sigma_\omega^2$$

denote the combined wedge between observed loss and chronic-poverty loss. If survey waves are not equally spaced, interpret σ_η^2 as the variance of chronic-consumption innovations per year and let t denote elapsed years since calibration.

Proposition 2 (Low-frequency benchmark dynamics identify W). Suppose the baseline observed-poverty benchmark is observed at calibration, at a first follow-up $t_1 > 0$, and at a later follow-up $t_2 > t_1$. Then

$$\sigma_\eta^2 = \frac{\mathcal{L}_{t_2}^{obs}(\tilde{c}_{i0}) - \mathcal{L}_{t_1}^{obs}(\tilde{c}_{i0})}{t_2 - t_1},$$

and

$$W = \frac{1}{2} [\mathcal{L}_{t_1}^{obs}(\tilde{c}_{i0}) - t_1\sigma_\eta^2].$$

Consequently, for any static targeting rule f and any horizon $t > 0$,

$$\mathcal{L}_t(f; c^P) = \mathcal{L}_t^{obs}(f) - W.$$

Proof. By Lemma 7, for every $t > 0$,

$$\mathcal{L}_t^{obs}(\tilde{c}_{i0}) = t\sigma_\eta^2 + 2W.$$

Taking the difference between the losses at t_2 and t_1 eliminates $2W$ and identifies σ_η^2 . Substituting the resulting slope into the expression for the first follow-up loss identifies W . The recovery of chronic-poverty loss follows from Corollary 3. $\square$

Corollary 8 (Application to PMTs). For PMTs and all horizons $t \geq 0$,

$$\mathcal{L}_t(\hat{c}_{i0}^{PMT}; c^P) = \mathcal{L}_t^{obs}(\hat{c}_{i0}^{PMT}) - W = t\sigma_\eta^2 + \sigma_\varepsilon^2.$$

A.7. High-frequency identification of measurement error

Low-frequency data identify only the combined wedge $W = \alpha^2\sigma_u^2 + \sigma_\omega^2$. To recover current-poverty loss, measurement error must be separated from transitory consumption. High-frequency data make this possible because measurement error creates an immediate jump in observed benchmark loss that true consumption cannot generate at the instant after baseline.

Proposition 3 (High-frequency data identify measurement error). Let

$$m(t) \equiv \mathbb{E}[(\tilde{c}_{it} - \tilde{c}_{i0})^2 \mid t]$$

denote the expected observed loss of the baseline observed-poverty benchmark at elapsed time $t \geq 0$ after baseline. Suppose

$$\lim_{t \downarrow 0} \mathbb{E}[(c_{it} - c_{i0})^2 \mid t] = 0, \tag{11}$$

so that true consumption does not jump discontinuously at the instant after baseline. Then

$$\Delta_0 \equiv \lim_{t \downarrow 0} [m(t) - m(0)] = 2\sigma_\omega^2. \tag{12}$$

Hence,

$$\sigma_\omega^2 = \frac{1}{2}\Delta_0. \tag{13}$$

Proof. Using $\tilde{c}_{it} = c_{it} + \omega_{it}$,

$$\begin{aligned} (\tilde{c}_{it} - \tilde{c}_{i0})^2 &= (c_{it} - c_{i0} + \omega_{it} - \omega_{i0})^2 \\ &= (c_{it} - c_{i0})^2 + (\omega_{it} - \omega_{i0})^2 + 2(c_{it} - c_{i0})(\omega_{it} - \omega_{i0}). \end{aligned}$$

Under classical measurement error, measurement errors are redrawn independently across surveys and are orthogonal to true consumption changes, so taking expectations conditional

on elapsed time $t > 0$ gives

$$m(t) = \mathbb{E}[(c_{it} - c_{i0})^2 \mid t] + 2\sigma_\omega^2.$$

At $t = 0$, $m(0) = 0$ mechanically because $(\tilde{c}_{i0} - \tilde{c}_{i0})^2 = 0$. Under condition (11),

$$\lim_{t \downarrow 0} \mathbb{E}[(c_{it} - c_{i0})^2 \mid t] = 0.$$

Therefore

$$\Delta_0 = \lim_{t \downarrow 0} [m(t) - m(0)] = 2\sigma_\omega^2,$$

which proves (12). Equation (13) follows immediately. $\square$

Corollary 9 (Recovering current-poverty loss and transitory poverty). Once σ_ω^2 is identified, current-poverty loss for any static targeting rule at horizons after calibration is

$$\mathcal{L}_t(f; c) = \mathcal{L}_t^{obs}(f) - \sigma_\omega^2, \quad (14)$$

and the transitory component is recovered as

$$\alpha^2 \sigma_u^2 = (\alpha^2 \sigma_u^2 + \sigma_\omega^2) - \sigma_\omega^2 = W - \sigma_\omega^2. \quad (15)$$

Proof. Equation (14) follows directly from Corollary 3. Equation (15) follows by subtracting the measurement-error variance identified in Proposition 3 from the combined wedge W identified in Proposition 2. $\square$

B. Data appendix

This appendix summarizes the data construction behind Section 3. The main sample requires repeated household consumption for at least three pre-Covid waves, PMT-style household characteristics, a representative or clearly defined study population, and enough observations to estimate targeting rules and evaluate them out of sample. The low-frequency analysis uses balanced analytical panels. The monthly India panel is used separately for high-frequency identification.

Table B.1 — Candidate household surveys

Survey family or candidate	Countries / examples	Main decision or exclusion reason
LSMS-ISA panels	Ethiopia, Malawi, Nigeria, Tanzania, Uganda	Included. These panels provide repeated consumption, PMT-style predictors, and at least three pre-Covid waves.
LSMS-related national panels	Ghana GSPS, Peru ENAHO, Vietnam VHLSS	Included. These surveys provide comparable repeated expenditure or consumption measures and usable household panels.
Comparable rural panels	Bangladesh BIHS, Thailand TVSEP	Included. These panels are not national LSMS surveys, but they provide repeated consumption and PMT predictors for clearly defined rural populations.
Two-wave LSMS or LSMS-style panels	Malawi IHPS 2010–2013, Niger ECVMA, Ethiopia ESS/ESPS Panel II	Excluded because only two usable rounds are available for the relevant panel window.
Older LSMS panels	Albania LSMS, Bosnia-Herzegovina LSMS, Nicaragua EMNV, Tanzania KHDS	Excluded because the panel period is too old for the main pre-Covid comparison.
LSMS cross-sections and repeated cross-sections	Ghana GLSS, Peru ENNIV, Nepal LSS, Tajikistan TLSS, Viet Nam VLSS/VHLSS 1992–2008, Malawi IHS	Excluded because they do not provide a household panel with at least three usable waves.
LSMS-supported high-frequency phone panels	Burkina Faso HFPS, Ethiopia HFPS, Malawi HFPS, Nigeria NLPS, Tanzania HFWMPs, Uganda HFPS	Excluded because they lack comparable household consumption data.
Middle- and higher-income non-LSMS panels	China CFPS, South Africa NIDS, Russia RLMS	Excluded because they represent higher- or middle-income contexts rather than the main low- and lower-middle-income comparison.
High-frequency non-LSMS panel	India CPHS	Excluded because existing assessments raise concerns about sample design, survey implementation, and under-representation of poor households.
Alternative Vietnam rural panel	UNU-WIDER VARHS	Excluded; the analysis instead uses VHLSS, the official national living-standards survey.

Note: Grouped rows list the closest included and excluded candidates rather than every survey fielded in each survey family.

Table B.2 — Panel construction and analytical sample sizes

Country	Waves	Effective starting HH	Linked balanced panel HH	Follow-up retention (%)	Analytical retention (%)	Final analytical HH
Bangladesh	3	5,503	5,503	100.0	91.0	5,010
Ethiopia	3	3,969	3,615	91.1	63.7	2,301
Ghana	3	5,009	4,016	80.2	88.5	3,553
Malawi	4	1,619	1,458	90.1	100.0	1,458
Nigeria	3	4,997	4,493	89.9	82.8	3,720
Peru	3	7,239	7,239	100.0	96.4	6,979
Tanzania	3	3,265	3,068	94.0	99.6	3,055
Thailand	6	2,186	1,793	82.0	70.8	1,269
Uganda	4	2,490	2,490	100.0	71.1	1,770
Vietnam	3	9,258	9,258	100.0	19.5	1,806
Total/mean		45,535	42,933	92.7	78.3	30,921

Note: Effective starting households are the relevant baseline or linked-panel denominator for the panel window used in the paper. Linked balanced panel households are observed in all retained waves after survey-specific identifier and lineage harmonization, but before welfare and predictor complete-case restrictions. Analytical retention is the final analytical sample divided by the linked balanced panel. The final analytical sample requires non-missing real per-capita consumption and PMT predictors in the retained waves.

Table B.3 — Consumption measures and price adjustment

Country	Welfare aggregate	Price basis	Main note
Bangladesh	Constructed annual household consumption	Official annual CPI	Rural BIHS panel.
Ethiopia	Survey-provided annual consumption aggregate	Official annual CPI	Rural and small-town panel.
Ghana	Constructed annual household consumption	Official annual CPI	GSPS panel aggregate harmonized across waves.
Malawi	Constructed annual household consumption	Official annual CPI	IHPS panel aggregate harmonized across waves.
Nigeria	Survey-provided annual consumption aggregate	Official annual CPI	Main analysis uses the post-harvest aggregate.
Peru	Survey-provided annual expenditure aggregate	Official annual CPI	ENAHO rotating panel.
Tanzania	Survey-provided annual consumption aggregate	Survey-specific real or price-adjusted basis	NPS panel.
Thailand	Survey-provided annual consumption aggregate	Survey-specific real or price-adjusted basis	Regional rural TVSEP panel.
Uganda	Survey-provided annual consumption aggregate	Survey-specific real or price-adjusted basis	UNPS post-refresh panel.
Vietnam	Constructed annual household expenditure	Official annual CPI	VHLSS rotating panel with expenditure module.
India	Monthly food consumption	Within-panel monthly values	Used for high-frequency identification rather than cross-country level comparisons.

Note: Low-frequency country-panel targeting exercises use real per-capita welfare and log consumption for mean-squared-error analysis. Where documented survey-provided real, PPP, or survey-specific price-adjusted aggregates are available, the analysis uses those sources and harmonizes price bases when necessary; otherwise it deflates the welfare aggregate used in the analysis with official CPI and expresses consumption in first-wave local-currency units. The India panel is used for within-panel high-frequency identification rather than cross-country level comparisons.

Table B.4 — Coverage of PMT predictor families

Country	No. of predictors	Geography	Household composition	Head characteristics	Housing	Consumer assets	Productive assets
Bangladesh	21	Yes	Yes	Yes	Yes	Yes	Partial
Ethiopia	68	Yes	Yes	Yes	Yes	Yes	Yes
Ghana	24	Yes	Yes	Yes	Yes	Yes	No
Malawi	24	Yes	Yes	Yes	Yes	Yes	No
Nigeria	52	Yes	Yes	Yes	Yes	Yes	No
Peru	23	Yes	Yes	Yes	Yes	Yes	No
Tanzania	74	Yes	Yes	Yes	Yes	Yes	Yes
Thailand	14	Yes	Partial	Yes	Yes	No	No
Uganda	43	Yes	Yes	Yes	Yes	Yes	Yes
Vietnam	57	Yes	Yes	Yes	Yes	Yes	Partial

Note: The table reports predictor families in the low-frequency country panels. The number of predictors counts predictor concepts rather than the number of estimated dummy-variable indicators: categorical dummies generated from one survey question count once, while distinct asset items count separately. Geography covers broad region, province, state, zone, rural, or urban indicators. Household composition covers household size and demographic structure. Head characteristics cover age, sex, marital status, literacy, and schooling. Housing covers dwelling materials, rooms, tenure, water, sanitation, electricity, lighting, and cooking fuel. Consumer assets cover household durables, communications, appliances, and transport assets. Productive assets cover land, livestock, farm equipment, business assets, and similar livelihood assets.

C. Additional results on poverty and PMT performance

This appendix reports the country-level poverty-dynamics and PMT-performance results that support Section 4. The main text reports the pooled patterns; the tables and figures below show the corresponding country-level inputs and make clear which numerical claims enter the decomposition.

C.1. Observed poverty dynamics

Table C.1 — Net poverty trends and gross household movement between survey waves

Country	Avg. years between waves	Net changes visible in repeated cross-sections				Gross household movement visible only in panels			
		Mean cons. change (% line)	Median cons. change (% line)	Poverty-rate change (pp)	Poverty-gap change (pp)	Mean abs. HH cons. change (% line)	Median abs. HH cons. change (% line)	Doubled or halved cons. (%)	Changed poverty status (%)
Bangladesh	3.5	+3.0	+3.2	-2.1	-0.4	71.7	36.7	14.6	26.3
Ethiopia	2.0	-24.6	-14.1	+6.9	+3.3	85.4	49.3	28.9	32.8
Ghana	4.0	+34.4	+29.8	-5.2	-1.3	219.3	124.7	44.9	24.0
Malawi	2.8	+10.1	+0.8	-0.5	-0.5	147.2	71.6	34.3	28.0
Nigeria	2.5	-11.9	-11.2	+4.1	+1.6	112.5	49.5	26.6	28.1
Peru	1.0	+4.1	+4.2	-1.1	-0.6	57.1	33.7	9.1	15.5
Tanzania	2.0	-10.5	-4.7	+2.0	+1.1	84.9	48.6	20.9	26.2
Thailand	2.0	+15.7	+10.9	-3.7	-1.1	101.7	57.4	21.8	22.7
Uganda	2.0	+17.8	+12.1	-3.7	-1.4	124.9	54.0	22.1	21.5
Vietnam	2.0	+13.6	+8.8	-3.6	-1.4	59.0	33.3	11.3	19.5
All countries	2.4	+5.2	+4.0	-0.7	-0.1	106.4	55.9	23.5	24.5

Note: Consumption is real per-capita welfare divided by the country-specific baseline 30th-percentile poverty line. Consumption changes are expressed as percent of that line; poverty-rate and poverty-gap changes are percentage points. The average-years column reports the average calendar time between adjacent survey waves. Entries are unweighted averages over adjacent survey-wave pairs and are not annualized. The All countries row is the unweighted mean across displayed country rows. Household-movement columns use the same adjacent wave pairs.

Table C.2 — Rank mobility and relative-poverty transitions between survey waves

Country	Avg. years between waves	Rank mobility				Relative-poverty transitions	
		Rank corr.	Mean abs. rank change	Median abs. rank change	Moved ≥ 20 rank points (%)	Entered/exited bottom 30 (%)	Bottom-30 persistence (%)
Bangladesh	3.5	0.55	21.0	16.6	42.2	25.9	56.9
Ethiopia	2.0	0.46	23.2	18.5	47.1	29.2	51.3
Ghana	4.0	0.47	23.1	18.5	46.7	29.4	50.9
Malawi	2.8	0.61	19.3	14.8	38.4	27.1	54.8
Nigeria	2.5	0.59	19.9	15.5	40.1	26.0	56.7
Peru	1.0	0.83	12.5	9.3	20.6	15.6	73.9
Tanzania	2.0	0.66	18.0	13.6	35.5	25.2	58.0
Thailand	2.0	0.54	21.1	16.7	42.7	27.6	54.0
Uganda	2.0	0.67	17.4	13.1	34.0	23.2	61.3
Vietnam	2.0	0.70	16.6	12.2	31.7	20.4	66.1
All countries	2.4	0.61	19.2	14.9	37.9	25.0	58.4

Note: Ranks are within-country, wave-specific percentile ranks of real per-capita welfare; lower ranks are poorer. Rank correlation is the Spearman correlation between household percentile ranks in adjacent waves. Absolute rank changes are measured in points on the 0–100 percentile-rank scale. Relative poverty is defined as being in the bottom 30 percent of the country-wave distribution. Entries are unweighted averages over adjacent survey-wave pairs and are not annualized. The All countries row is the unweighted mean across displayed country rows. Bottom-30 persistence is the share of households in the bottom 30 percent in one wave that remain there in the next wave.

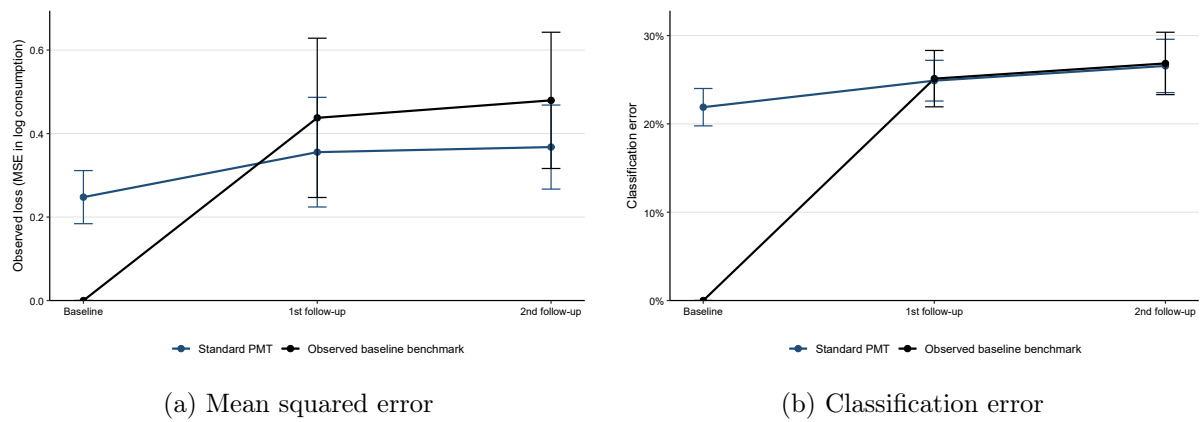
Figure C.1 — Decile transition matrices



Notes: Rows show the previous-wave decile and columns show the next-wave decile, with decile 1 the poorest and decile 10 the richest. Cells report transition probabilities, in percent, pooled over adjacent survey-wave pairs within each country. Darker cells indicate more frequent transitions. Concentration along the diagonal indicates relative-rank persistence; mass far from the diagonal indicates larger rank reshuffling.

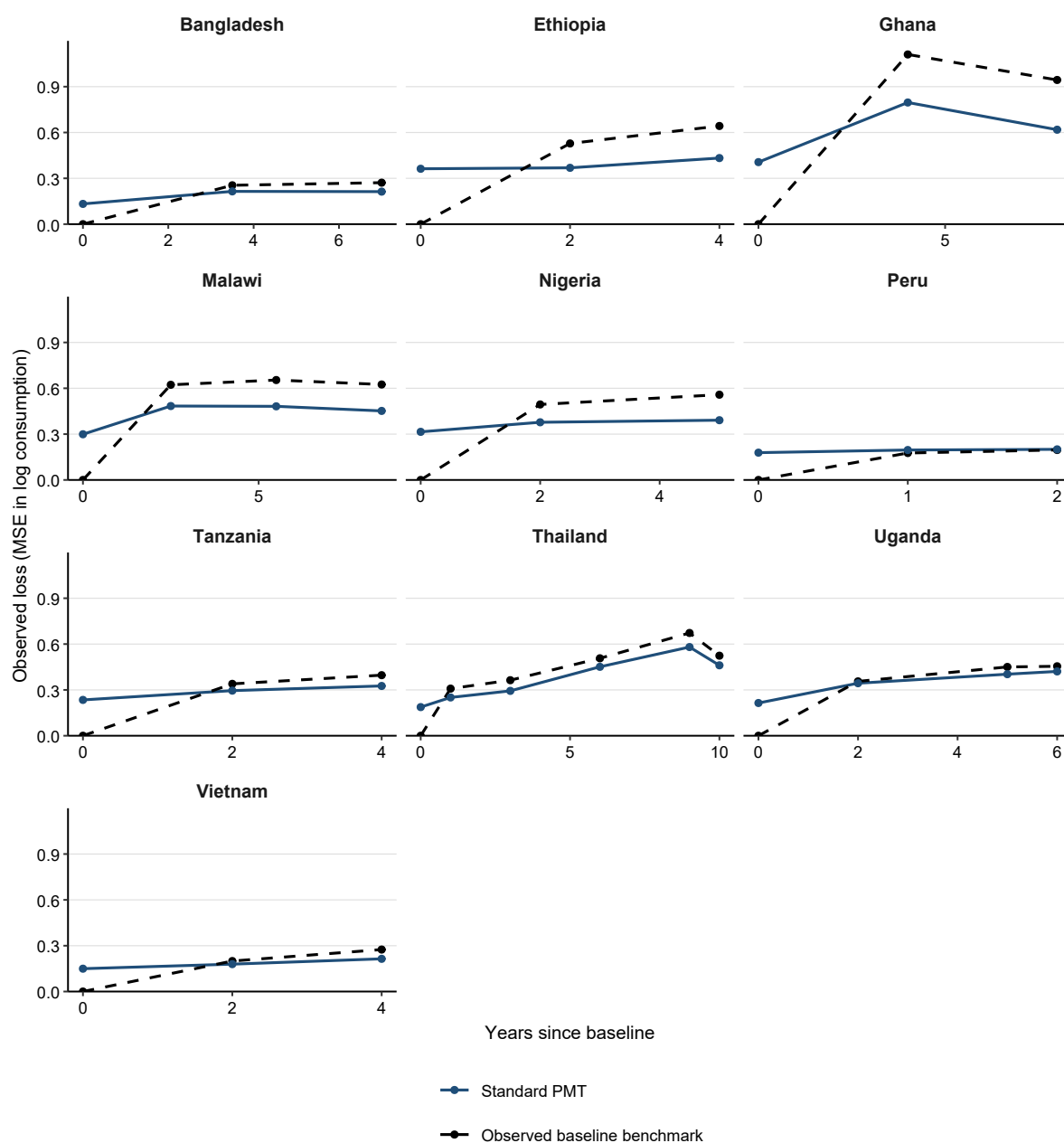
C.2. Observed targeting performance over time

Figure C.2—Observed targeting performance over time, with confidence intervals



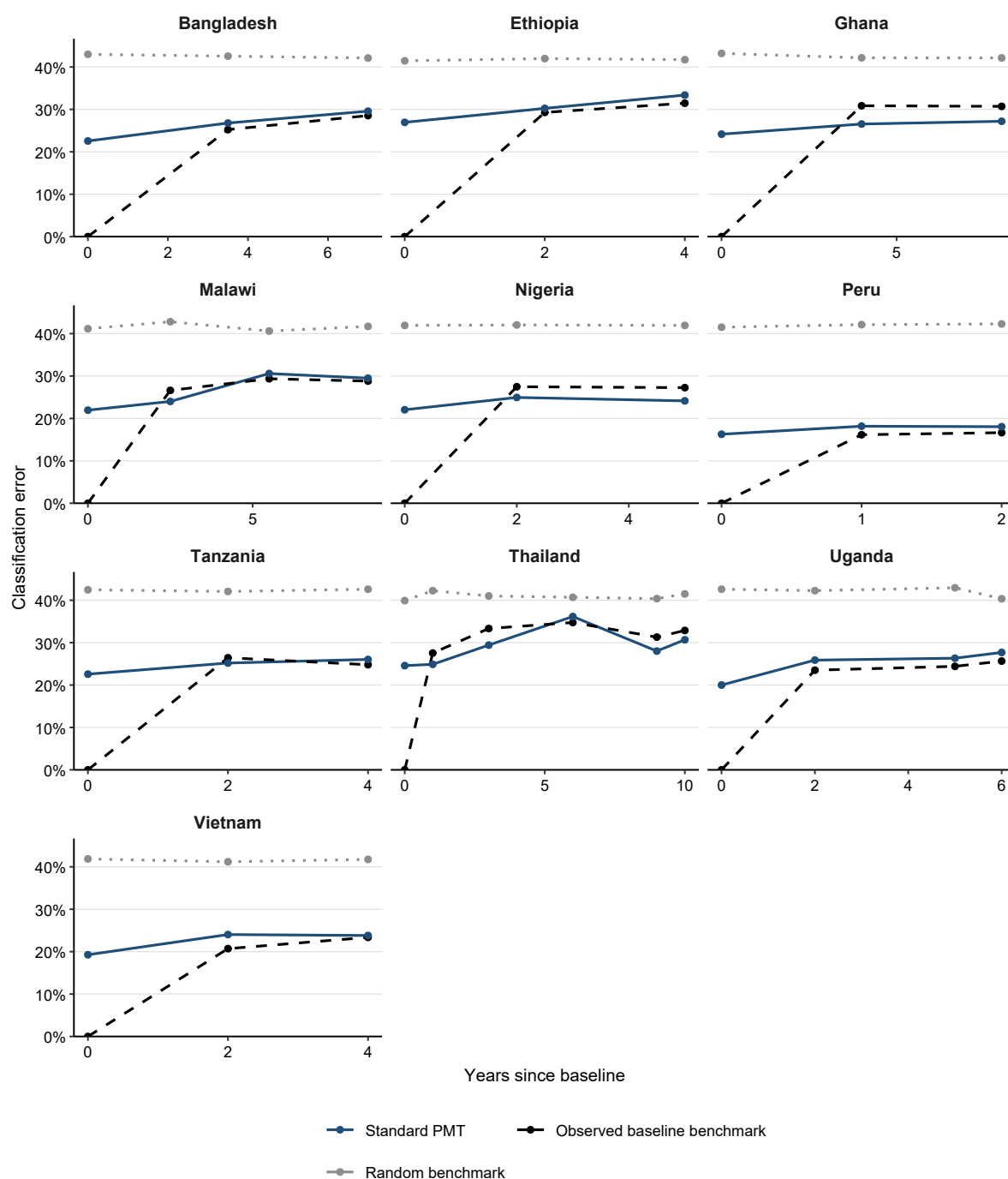
Notes: The figure repeats Figure 3 with 95% confidence intervals across country panels, after averaging over the 100 repeated train-test splits within each country.

Figure C.3 — Observed targeting performance by country: mean squared error



Notes: The figure reports observed mean squared error over time for the standard PMT and the baseline observed-poverty benchmark, separately by country. The PMT is calibrated at baseline and evaluated out of sample in later waves. Values are averaged across 100 repeated train-test splits.

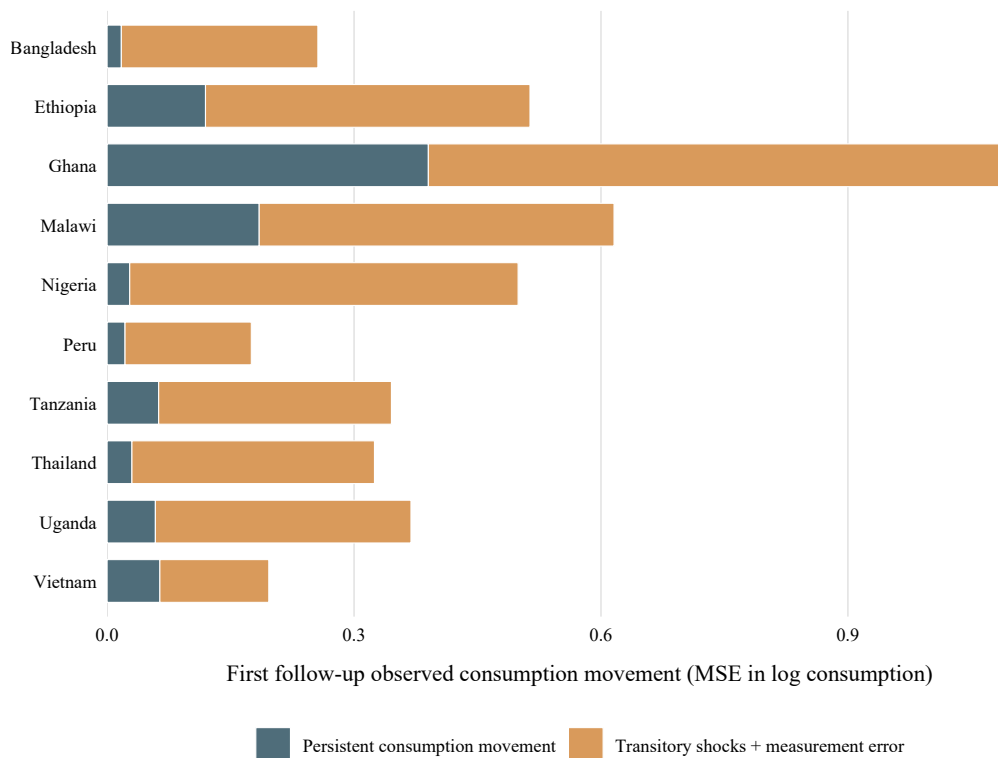
Figure C.4— Observed targeting performance by country: classification error



Notes: The figure reports bottom-30 classification error over time for the standard PMT and the baseline observed-poverty benchmark, separately by country. The PMT is calibrated at baseline and evaluated out of sample in later waves. Values are averaged across 100 repeated train-test splits.

C.3. Decomposition inputs

Figure C.5 — Observed consumption movement by component



Notes: The figure decomposes first follow-up within-household movement in log real per-capita consumption. For most countries, the persistent component is inferred from the post-baseline observed-consumption benchmark path. For Ghana and Malawi, where the observed benchmark and PMT both contain large wave-specific movements, the persistent component is recovered from the contemporaneous increase in standard PMT loss. The transitory-shocks plus measurement-error component is the residual full two-period contribution to observed movement, the $2W$ object. Low-frequency panels identify this combined component but do not separately identify its two parts.

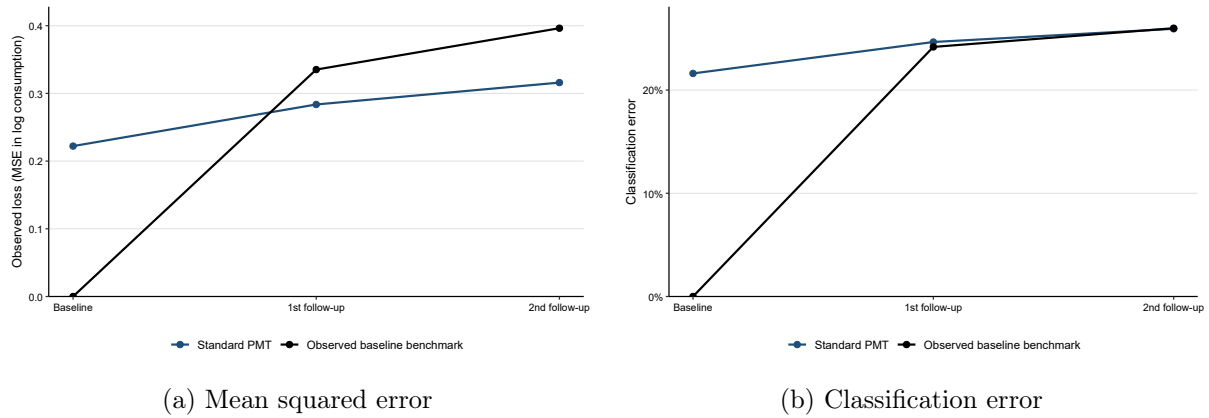
Table C.3 — Model decomposition inputs for observed consumption movement

Country	Waves	Span (years)	First gap (years)	Initial loss	Persistent part	Trans./ME part	Persistent share (%)	Trans./ME share (%)
Bangladesh	3	7.0	3.5	0.256	0.017	0.239	6.7	93.3
Ethiopia	3	4.0	2.0	0.514	0.120	0.394	23.3	76.7
Ghana	3	8.0	4.0	1.084	0.390	0.694	36.0	64.0
Malawi	4	8.5	2.5	0.616	0.185	0.431	30.0	70.0
Nigeria	3	5.0	2.0	0.499	0.027	0.472	5.5	94.5
Peru	3	2.0	1.0	0.175	0.022	0.153	12.4	87.6
Tanzania	3	4.0	2.0	0.345	0.063	0.283	18.1	81.9
Thailand	6	10.0	1.0	0.325	0.030	0.295	9.3	90.7
Uganda	4	6.0	2.0	0.369	0.059	0.311	15.9	84.1
Vietnam	3	4.0	2.0	0.196	0.064	0.132	32.6	67.4
All countries	3.5	5.8	2.2	0.438	0.098	0.340	19.0	81.0

Note: Losses are mean squared errors in log real per-capita consumption. Initial loss is the first follow-up loss of targeting households by baseline observed consumption. The persistent part is the implied chronic-consumption movement over the first interval, estimated from the post-baseline benchmark path except in Ghana and Malawi, where it is recovered from the contemporaneous increase in standard PMT loss. The transitory/ME part is the remaining part of the initial loss. Component shares divide each component by the initial loss. All countries is the unweighted mean across country rows.

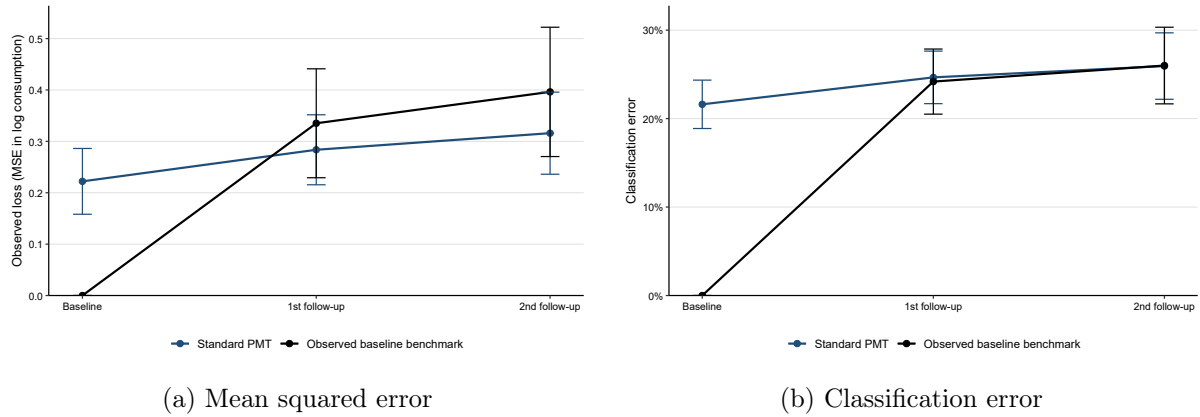
C.4. Robustness excluding Ghana and Malawi

Figure C.6 — Observed targeting performance excluding Ghana and Malawi



Notes: The figure repeats the main observed targeting-performance figure after excluding Ghana and Malawi, the two country panels with the largest wave-specific movements in the decomposition exercise. Values are averaged across the remaining eight low-frequency country panels and 100 repeated train-test splits.

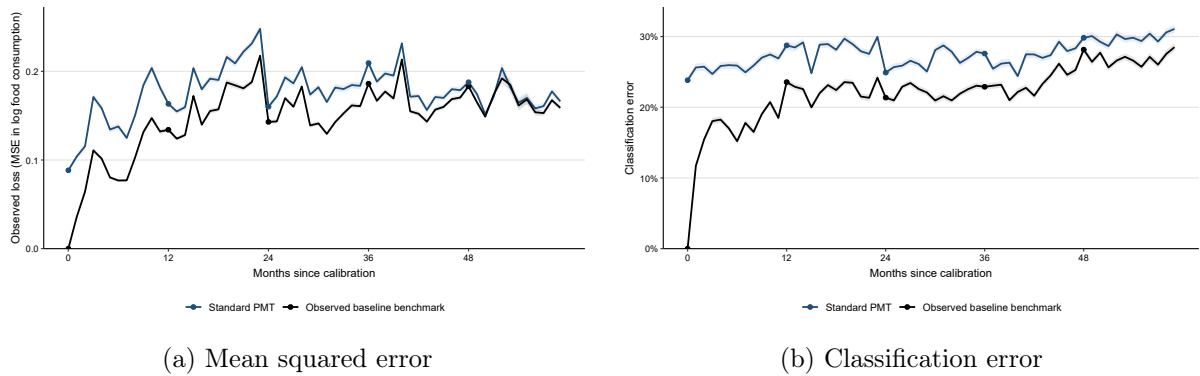
Figure C.7— Observed targeting performance excluding Ghana and Malawi, with confidence intervals



Notes: The figure repeats Appendix Figure C.6 with 95% confidence intervals across country panels, after averaging over the 100 repeated train-test splits within each country.

C.5. Monthly India targeting performance over the full panel

Figure C.8— Observed benchmark deterioration at monthly frequency: full panel



Notes: The figure repeats Figure 4 over the full 60-month India panel. Panel (a) reports mean squared error in log food consumption. Panel (b) reports bottom-30 classification error. Values are averaged across 100 repeated train-test splits; shaded bands report 95% confidence intervals across repeated splits.

C.6. PMT loss components

The next tables and figures support the PMT-performance results in Section 4. They report the country-level MSE decomposition, the final-wave classification-error increase, and two checks showing that the PMT result is not driven only by households close to the poverty line.

Table C.4 — PMT MSE loss components and annual increases over time

Country	Observed loss (L_0^{obs})	Transitory and measurement wedge ($\alpha^2 \sigma_u^2 + \sigma_w^2$)	Chronic mistargeting (σ_ϵ^2)	Change in chronic poverty per year (σ_η^2)	PMT error increase per year ($\Delta L_{0T}^{obs}/T$)
Bangladesh	0.133	0.119 (90%)	0.013 (10%)	0.005 (4%)	0.011 (9%)
Ethiopia	0.363	0.197 (54%)	0.165 (46%)	0.060 (16%)	0.018 (5%)
Ghana	0.406	0.356 (88%)	0.050 (12%)	0.027 (7%)	0.027 (7%)
Malawi	0.299	0.236 (79%)	0.063 (21%)	0.018 (6%)	0.018 (6%)
Nigeria	0.315	0.236 (75%)	0.079 (25%)	0.014 (4%)	0.015 (5%)
Peru	0.179	0.077 (43%)	0.102 (57%)	0.022 (12%)	0.011 (6%)
Tanzania	0.235	0.141 (60%)	0.093 (40%)	0.031 (13%)	0.023 (10%)
Thailand	0.187	0.147 (79%)	0.040 (21%)	0.030 (16%)	0.027 (15%)
Uganda	0.214	0.155 (72%)	0.059 (28%)	0.029 (14%)	0.034 (16%)
Vietnam	0.150	0.066 (44%)	0.083 (56%)	0.032 (21%)	0.016 (11%)
Average	0.248	0.173 (70%)	0.075 (30%)	0.027 (11%)	0.020 (8%)

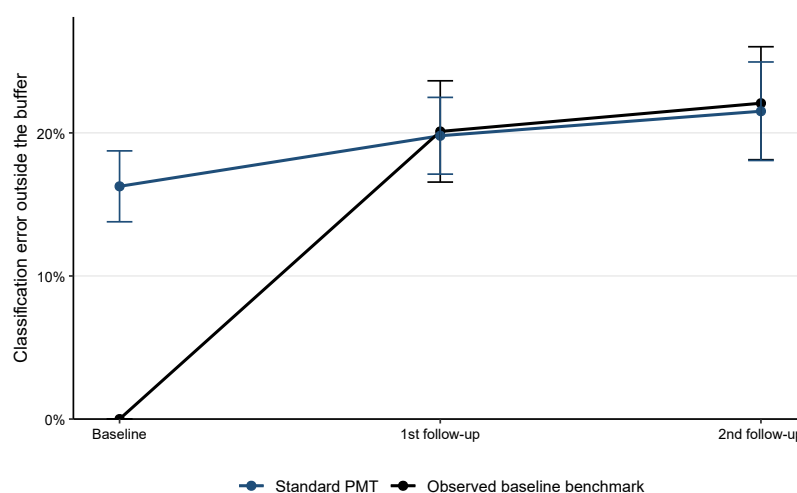
Note: The table decomposes baseline PMT loss. Observed loss, L_0^{obs} , is the mean squared error between the baseline OLS-based PMT score and observed log real per-capita consumption at calibration. Let $\mathcal{L}_t^{obs}(\tilde{c}_{i0})$ denote the observed loss of the baseline observed-consumption benchmark t years after calibration, let t_1 denote the first follow-up year, and let T denote the final observed year. For most countries, change in chronic poverty, σ_η^2 , is estimated as $(\mathcal{L}_T^{obs}(\tilde{c}_{i0}) - \mathcal{L}_{t_1}^{obs}(\tilde{c}_{i0})) / (T - t_1)$, and the transitory and measurement wedge is $W = (\mathcal{L}_{t_1}^{obs}(\tilde{c}_{i0}) - t_1 \sigma_\eta^2) / 2$. For Ghana and Malawi, where the observed benchmark path contains large wave-specific movements, the chronic component is recovered from the contemporaneous increase in standard PMT loss; W is the average across follow-ups of one half of observed benchmark loss net of the PMT loss increase. Chronic mistargeting, σ_ϵ^2 , is observed loss minus W ; it captures baseline PMT error against chronic consumption rather than noisy observed consumption. PMT error increase per year, $\Delta L_{0T}^{obs}/T$, is the increase in observed PMT loss between baseline and the final observed wave divided by years since baseline. Percentages in parentheses express each entry as a share of baseline observed loss, L_0^{obs} . The average row is an unweighted average across country rows; its percentages are computed from the average levels.

Table C.5 — Increase in standard PMT classification error over time

Country	Baseline error (%)	Final-wave error (%)	Years to final wave	Increase (pp)	Annualized increase (pp/year)
Bangladesh	22.6	29.6	7.0	+7.0	+1.00
Ethiopia	26.9	33.4	4.0	+6.4	+1.61
Ghana	24.2	27.2	8.0	+3.0	+0.38
Malawi	21.9	29.5	8.5	+7.5	+0.89
Nigeria	22.0	24.1	5.0	+2.1	+0.42
Peru	16.3	18.0	2.0	+1.8	+0.89
Tanzania	22.6	26.0	4.0	+3.5	+0.87
Thailand	24.6	30.7	10.0	+6.1	+0.61
Uganda	20.0	27.7	6.0	+7.7	+1.28
Vietnam	19.3	23.8	4.0	+4.5	+1.14
All countries	22.0	27.0	5.9	+5.0	+0.91

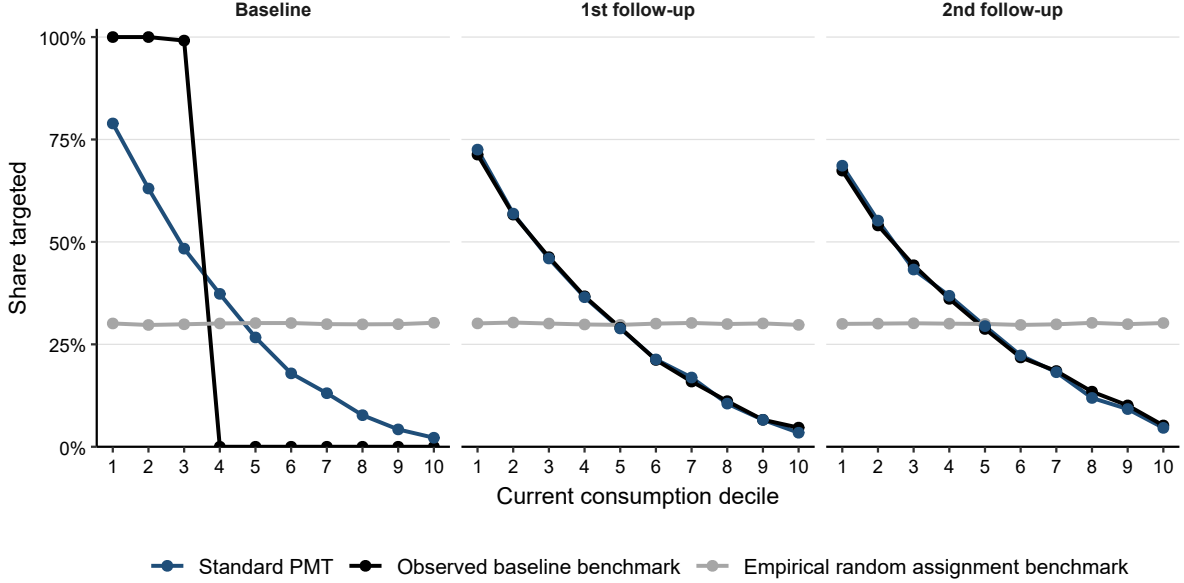
Note: The table reports total classification error for the standard PMT targeting the bottom 30 percent of the country-wave welfare distribution. The PMT is trained at baseline and evaluated in the final available wave for each country. Classification error is the share of held-out households whose predicted bottom-30 status differs from observed bottom-30 status. The All countries row is the unweighted mean across country rows; its annualized increase is the mean of country-specific annualized increases.

Figure C.9 — Classification error outside a buffer around the poverty line



Notes: The figure reports bottom-30 classification error after excluding households whose current observed consumption rank lies within 10 percentile points on either side of the poverty line. The PMT and observed-poverty benchmark are calibrated at baseline and evaluated out of sample over time. Values are averaged across the ten low-frequency country panels and 100 repeated train-test splits; error bars report 95% confidence intervals across country panels, after averaging over repeated splits within each country.

Figure C.10 — Targeting across the consumption distribution with random assignment benchmark



Notes: The figure reports the share of households targeted by each rule within each current observed-consumption decile. Decile 1 is the poorest and decile 10 the richest in the evaluation wave. The observed-poverty benchmark is shown as a solid black line. The gray line reports empirical random assignment, generated by assigning a random score within each held-out country split and targeting the lowest 30% of that random score distribution. Values are averaged across the ten low-frequency country panels and 100 repeated train-test splits.

D. Targeting design appendix

This appendix formalizes what the targeting designs considered in Section 5 can and cannot improve in the framework of Section 2.3. Throughout, current consumption is

$$c_{it} = p_{it} + \alpha u_{it},$$

permanent income evolves as

$$p_{it} = p_{i,t-1} + \eta_{it},$$

and the contemporaneous linear projection of permanent income on observed characteristics is

$$p_{it} = \beta_t \Gamma_{it} + \varepsilon_{it}, \quad \text{Cov}(\Gamma_{it}, \varepsilon_{it}) = 0.$$

Observed consumption is

$$\tilde{c}_{it} = c_{it} + \omega_{it} = p_{it} + \alpha u_{it} + \omega_{it}.$$

D.1. Derivations for static refinements

Any static PMT score is fixed at baseline. It can therefore improve targeting only by reducing baseline chronic-poverty prediction error. It cannot eliminate transitory poverty at the time of evaluation, and it cannot eliminate permanent innovations that arrive after targeting.

Proposition 4 (Static prediction only affects baseline chronic-poverty mistargeting). Let $\hat{c}_{i0}$ be any static targeting score measurable with respect to baseline observables. Then true loss against current consumption at horizon t is

$$\mathcal{L}_t(\hat{c}_{i0}; c) = t\sigma_\eta^2 + \alpha^2\sigma_u^2 + \mathbb{E}[(p_{i0} - \hat{c}_{i0})^2]. \quad (16)$$

Equivalently, true loss against chronic poverty is

$$\mathcal{L}_t(\hat{c}_{i0}; c^P) = t\sigma_\eta^2 + \mathbb{E}[(p_{i0} - \hat{c}_{i0})^2].$$

Thus any static refinement can improve performance only through the permanent-consumption prediction error $\mathbb{E}[(p_{i0} - \hat{c}_{i0})^2]$.

Proof. Using $c_{it} = p_{i0} + \sum_{s=1}^t \eta_{is} + \alpha u_{it}$,

$$c_{it} - \hat{c}_{i0} = (p_{i0} - \hat{c}_{i0}) + \sum_{s=1}^t \eta_{is} + \alpha u_{it}.$$

Squaring and taking expectations, the cross terms vanish by orthogonality of $(p_{i0} - \hat{c}_{i0})$, permanent innovations, and transitory shocks. Hence

$$\mathbb{E}[(c_{it} - \hat{c}_{i0})^2] = \mathbb{E}[(p_{i0} - \hat{c}_{i0})^2] + \text{Var}\left(\sum_{s=1}^t \eta_{is}\right) + \alpha^2 \text{Var}(u_{it}),$$

which yields (16). The chronic-poverty result follows by replacing c_{it} with $c_{it}^P = p_{it}$. $\square$

Proposition 5 (Machine learning can only reduce baseline chronic-poverty prediction error). Let the standard PMT score be $\hat{c}_{i0}^{PMT} = \beta\Gamma_{i0}$, and let a machine-learning PMT score be $\hat{c}_{i0}^{ML} = m(\Gamma_{i0})$. Then

$$\mathcal{L}_t(\hat{c}_{i0}^{PMT}; c) - \mathcal{L}_t(\hat{c}_{i0}^{ML}; c) = \mathbb{E}[(p_{i0} - \beta\Gamma_{i0})^2] - \mathbb{E}[(p_{i0} - m(\Gamma_{i0}))^2].$$

The same equality holds for chronic-poverty loss. Thus machine learning improves static PMT performance if and only if it reduces baseline permanent-consumption prediction error.

Proof. Apply Proposition 4 to $\hat{c}_{i0}^{PMT}$ and $\hat{c}_{i0}^{ML}$ and subtract. $\square$

Proposition 6 (Averaging the PMT training target does not change the population signal). Suppose the PMT is trained on an average of observed consumption over several

pre-targeting observations,

$$\bar{\tilde{c}}_i = \frac{1}{K} \sum_{k=1}^K \tilde{c}_{i\tau_k},$$

using baseline characteristics Γ_{i0} . Let $p_{i0} = \beta\Gamma_{i0} + \varepsilon_{i0}$. For each pre-targeting observation, write

$$\tilde{c}_{i\tau_k} = p_{i0} + \nu_{i\tau_k} + \alpha u_{i\tau_k} + \omega_{i\tau_k},$$

where $\nu_{i\tau_k}$ captures the difference between permanent income in period τ_k and baseline permanent income. If $\nu_{i\tau_k}$, $u_{i\tau_k}$, and $\omega_{i\tau_k}$ are orthogonal to Γ_{i0} , then the population slope from regressing $\bar{\tilde{c}}_i$ on Γ_{i0} is β . The averaged-target PMT therefore has the same structural chronic-poverty loss as the standard PMT:

$$\mathcal{L}_t(\hat{c}_{i0}; c^P) = t\sigma_\eta^2 + \sigma_\varepsilon^2,$$

where $\sigma_\varepsilon^2 = \text{Var}(\varepsilon_{i0})$. Hence averaging the training target does not reduce the structural chronic-poverty mistargeting component.

Proof. The population slope from the averaged-target regression is

$$\beta^{avg} = \frac{\text{Cov}(\bar{\tilde{c}}_i, \Gamma_{i0})}{\text{Var}(\Gamma_{i0})}.$$

Using the averaged target,

$$\text{Cov}(\bar{\tilde{c}}_i, \Gamma_{i0}) = \text{Cov}(p_{i0}, \Gamma_{i0}) + \frac{1}{K} \sum_{k=1}^K [\text{Cov}(\nu_{i\tau_k}, \Gamma_{i0}) + \alpha \text{Cov}(u_{i\tau_k}, \Gamma_{i0}) + \text{Cov}(\omega_{i\tau_k}, \Gamma_{i0})].$$

The orthogonality assumptions set the bracketed terms to zero. Since $p_{i0} = \beta\Gamma_{i0} + \varepsilon_{i0}$ and $\text{Cov}(\Gamma_{i0}, \varepsilon_{i0}) = 0$, $\text{Cov}(p_{i0}, \Gamma_{i0}) = \beta \text{Var}(\Gamma_{i0})$, so $\beta^{avg} = \beta$. Therefore the averaged-target score is $\hat{c}_{i0} = \beta\Gamma_{i0}$, and

$$p_{it} - \hat{c}_{i0} = \left(p_{i0} + \sum_{s=1}^t \eta_{is} \right) - \beta\Gamma_{i0} = \varepsilon_{i0} + \sum_{s=1}^t \eta_{is}.$$

Squaring and taking expectations yields

$$\mathcal{L}_t(\hat{c}_{i0}; c^P) = \text{Var}(\varepsilon_{i0}) + \text{Var}\left(\sum_{s=1}^t \eta_{is}\right) = \sigma_\varepsilon^2 + t\sigma_\eta^2.$$

The way the dependent variable is averaged in the training regression therefore does not alter this population decomposition, although it may matter in finite samples. $\square$

Proposition 7 (Forecasting PMTs still target the permanent component encoded in lagged observables). Consider a forecasting PMT estimated by regressing future observed consumption $\tilde{c}_{it}$ on lagged characteristics $\Gamma_{i,t-1}$:

$$\tilde{c}_{it} = \beta_0^F + \beta^F \Gamma_{i,t-1} + e_{it}^F.$$

Then

$$\beta^F = \frac{\text{Cov}(\tilde{c}_{it}, \Gamma_{i,t-1})}{\text{Var}(\Gamma_{i,t-1})} = \frac{\text{Cov}(p_{it}, \Gamma_{i,t-1})}{\text{Var}(\Gamma_{i,t-1})}.$$

If η_{it} , u_{it} , and ω_{it} are orthogonal to $\Gamma_{i,t-1}$, then

$$\text{Cov}(p_{it}, \Gamma_{i,t-1}) = \text{Cov}(p_{i,t-1}, \Gamma_{i,t-1}).$$

Thus a forecasting PMT still identifies the permanent component encoded in lagged observables. It cannot predict future transitory shocks, future measurement error, or future permanent innovations.

Proof. From $\tilde{c}_{it} = p_{it} + \alpha u_{it} + \omega_{it}$,

$$\text{Cov}(\tilde{c}_{it}, \Gamma_{i,t-1}) = \text{Cov}(p_{it}, \Gamma_{i,t-1}) + \alpha \text{Cov}(u_{it}, \Gamma_{i,t-1}) + \text{Cov}(\omega_{it}, \Gamma_{i,t-1}).$$

Orthogonality implies the last two terms are zero. Next, $p_{it} = p_{i,t-1} + \eta_{it}$, so

$$\text{Cov}(p_{it}, \Gamma_{i,t-1}) = \text{Cov}(p_{i,t-1}, \Gamma_{i,t-1}) + \text{Cov}(\eta_{it}, \Gamma_{i,t-1}).$$

Orthogonality of η_{it} and $\Gamma_{i,t-1}$ yields the claim. $\square$

D.2. Derivations for updating PMTs over time

We now compare three updating strategies. Write the contemporaneous projection of permanent income as

$$p_{it} = \beta_t \Gamma_{it} + \varepsilon_{it}.$$

Proposition 8 (Updating predictors only). Suppose the PMT function is held fixed at its baseline value β_0 , but population characteristics are updated over time, so the targeting score is

$$\hat{c}_{it}^{char} = \beta_0 \Gamma_{it}.$$

Then chronic-poverty loss is

$$\mathcal{L}_t(\hat{c}_{it}^{char}; c^P) = \mathbb{E}[(\varepsilon_{it} + (\beta_t - \beta_0)\Gamma_{it})^2],$$

and current-poverty loss adds the transitory component:

$$\mathcal{L}_t(\hat{c}_{it}^{char}; c) = \mathbb{E}[(\varepsilon_{it} + (\beta_t - \beta_0)\Gamma_{it})^2] + \alpha^2 \sigma_u^2.$$

Thus updating predictors helps if characteristics track chronic poverty and the mapping from characteristics to chronic poverty is sufficiently stable over time.

Proof. Using $p_{it} = \beta_t \Gamma_{it} + \varepsilon_{it}$,

$$p_{it} - \hat{c}_{it}^{char} = (\beta_t \Gamma_{it} + \varepsilon_{it}) - \beta_0 \Gamma_{it} = \varepsilon_{it} + (\beta_t - \beta_0)\Gamma_{it}.$$

Squaring and taking expectations yields the chronic-poverty expression. Current-poverty loss adds $\alpha^2 \sigma_u^2$. $\square$

Proposition 9 (Updating the function only). Suppose the PMT function is re-estimated each period, but population characteristics remain stale at their baseline values, so the targeting score is

$$\hat{c}_{it}^{fun} = \beta_t \Gamma_{i0}.$$

Then chronic-poverty loss is

$$\mathcal{L}_t(\hat{c}_{it}^{fun}; c^P) = \mathbb{E}[(\varepsilon_{it} + \beta_t(\Gamma_{it} - \Gamma_{i0}))^2],$$

and current-poverty loss again adds $\alpha^2 \sigma_u^2$. Thus updating the function alone cannot eliminate the error caused by stale population characteristics.

Proof. Using $p_{it} = \beta_t \Gamma_{it} + \varepsilon_{it}$,

$$p_{it} - \hat{c}_{it}^{fun} = (\beta_t \Gamma_{it} + \varepsilon_{it}) - \beta_t \Gamma_{i0} = \varepsilon_{it} + \beta_t(\Gamma_{it} - \Gamma_{i0}).$$

The result follows by squaring and taking expectations. $\square$

Proposition 10 (Updating both the function and predictors). Suppose both the PMT function and population characteristics are updated, so the targeting score is

$$\hat{c}_{it}^{both} = \beta_t \Gamma_{it}.$$

Then chronic-poverty loss is

$$\mathcal{L}_t(\hat{c}_{it}^{both}; c^P) = \mathbb{E}[\varepsilon_{it}^2],$$

and current-poverty loss is

$$\mathcal{L}_t(\hat{c}_{it}^{both}; c) = \mathbb{E}[\varepsilon_{it}^2] + \alpha^2 \sigma_u^2.$$

Thus updating both the function and predictors is the only PMT design that removes the mechanical deterioration due to predictable changes in chronic poverty.

Proof. Immediate from

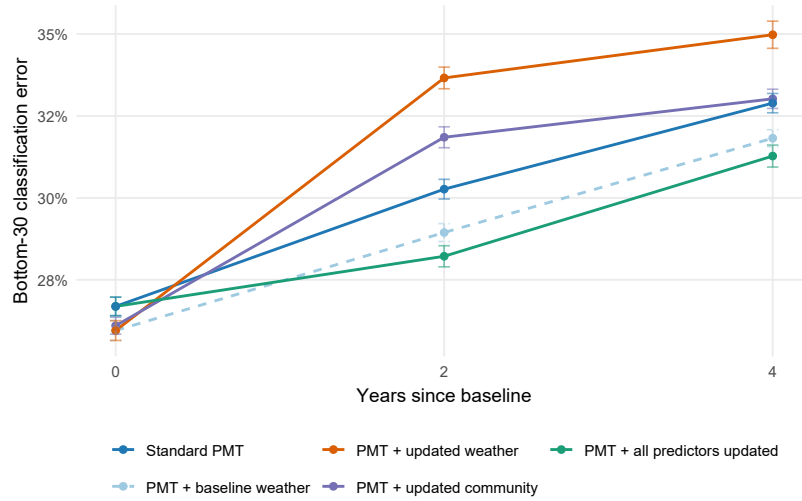
$$p_{it} - \hat{c}_{it}^{both} = (\beta_t \Gamma_{it} + \varepsilon_{it}) - \beta_t \Gamma_{it} = \varepsilon_{it}.$$

Current-poverty loss adds the transitory term $\alpha^2 \sigma_u^2$. $\square$

Propositions 8–10 clarify the empirical results in Section 5.2. Updating predictors alone can help partially if observables track chronic poverty. Updating the function alone has limited scope when population characteristics are stale. Updating both is the only PMT design that can systematically keep up with predictable changes in chronic poverty, though it remains unable to track transitory poverty with a static cross-sectional score.

D.3. Weather and community updates in Ethiopia

Figure D.1 — Updating PMTs with weather and community information in Ethiopia



Notes: The figure reports out-of-sample classification error in the Ethiopia panel. The standard PMT uses baseline household predictors. The baseline-weather specification adds baseline meteorological predictors and keeps them fixed. The updated-weather and updated-community specifications keep the PMT formula and baseline household predictors fixed but refresh the indicated low-cost information set. The all-predictors-updated specification refreshes the ordinary PMT predictors while keeping the baseline formula fixed. Values are averaged across 100 repeated train-test splits; error bars report 95% confidence intervals across repeated splits.